Scaling limits and surveillance design for epidemic reproduction numbers under environmental and reporting stochasticity

Apevon Science

Abstract

Estimates of the reproduction number from surveillance counts are usually assumed to become more precise as more cases are reported, although transmission varies between periods. Using a branching process with environmental variation in transmission, negative binomial offspring and binomial reporting, we show that the error variance of the mean reproduction number per period is $V(m) = v_R + A/m$ for m reported cases. Precision gains therefore diminish beyond the balance scale $m_\times = A/v_R$, and the budget-optimal case volume is $m^* = \sqrt{(c_0/c_1)m_\times}$ per period for a cost ratio c_0/c_1 of periods to cases. We analysed US influenza hospital admissions (10506 state-weeks, 2020–2026). In the early-growth windows of four seasons, the relative loss of a renewal-type ratio declined with admissions towards a positive intercept. This gave a balance scale of 37.4 admissions per week on the scale of that ratio (95% CI: 19.1, 71.0). A two-step inversion through data-matched simulations mapped this ratio and a weighted growth ratio to latent environmental variances of 0.040 (95% CI: 0.026, 0.056) and 0.043 (95% CI: 0.025, 0.064). The two estimates were consistent (ratio 0.93). Both are conditional on a 3.6-day generation interval and on independent weekly shocks, as the lag-1 autocorrelation of the data supported. Of the intercept, 41% to 51% was common to all states in a calendar week. Under mandatory reporting, facility completeness changed the intercept by 2% (upper bound 5.3%; equivalence margin 17% at reference volatility 0.10). Weekly growth was 1.58 times (95% CI: 0.95, 2.23) as dispersed in the 2024 voluntary period as in the same months of 2023. Where weekly admissions exceed the balance scale, as in 55% of state-weeks in the growth windows, further periods and regions improve precision more than further cases per period. High facility participation was associated with lower volatility.

Keywords:
Reproduction number
Environmental stochasticity
Surveillance design
Reporting heterogeneity
Influenza hospitalisation
Diminishing returns

1. Introduction

Estimates of the time-varying reproduction number have become a routine output of infectious disease surveillance. They were used to track the transmission of Ebola virus disease in West Africa (Nouvellet et al., 2018), to assess non-pharmaceutical interventions against COVID-19 (Flaxman et al., 2020) and to describe the early spread of SARS-CoV-2 (Kucharski et al., 2020). They are now produced routinely for many countries (Huisman et al., 2022), with practical guidance on their estimation (Gostic et al., 2020; Steyn et al., 2025). Their value for situational awareness depends on how precisely they can be estimated from the data that surveillance systems actually collect.

The instantaneous reproduction number, R_t, is the expected number of secondary infections generated by an individual infected at time t if conditions remained as they were at that time (Fraser, 2007). It is usually estimated from incidence through the renewal equation, which weights past incidence by the generation interval (the time between infection of a primary case and infection of its secondary cases) (Wallinga and Teunis, 2004; Cori et al., 2013; Thompson et al., 2019). For a given generation interval, the posterior uncertainty of such estimates shrinks as the number of cases in the estimation window increases (Cori et al., 2013). Parag and colleagues quantified the information that incidence carries about R_t in closed form, including the loss caused by time-varying reporting fractions, and derived the limits it places on timely decisions (Parag, 2021; Parag et al., 2022, 2025). Extensions handle temporally aggregated counts (Nash et al., 2023; Ogi-Gittins et al., 2024) and the difference between the reproduction number of infections and the apparent reproduction number of a delayed observation series (Eales and Riley, 2024).

These precision results measure error against the realised value of R_t in the period being estimated, and that error vanishes as case counts grow. Many uses of R_t estimates refer instead to a quantity that persists across periods, such as the typical level of transmission in a season or the value expected next week. Policy evaluations likewise average over weeks with different weather, school calendars and contact patterns (Dalziel et al., 2018). Between-period variation in R_t is shared by all chains of transmission in a population at a given time. Population ecology distinguishes such environmental stochasticity from demographic stochasticity, which arises from independent individual outcomes (Engen et al., 1998; Rohani et al., 2002; Lande et al., 2003), and mechanistic epidemic models have included both forms (Bretó et al., 2009). Demographic variance per individual falls with population size, whereas environmental variance does not,

and the population size at which the two contributions are equal is a standard summary of extinction risk (Lande et al., 2003). Overdispersed offspring distributions (Lloyd-Smith et al., 2005; Endo et al., 2020) and incomplete reporting add to the demographic side of this partition.

Surveillance data add a third source of variation. The proportion of infections that appear in a count changes over time with testing practice, care seeking and the completeness of facility reporting (Azmon et al., 2014; Stoner et al., 2019; Parag and Donnelly, 2022). In the United States, reporting of influenza hospital admissions to the National Healthcare Safety Network (NHSN) was mandatory until April 2024, voluntary from May to October 2024 and mandatory again from November 2024 (Centers for Disease Control and Prevention, 2026). Hospital admission series of this kind are increasingly used for forecasting and real-time assessment (Mathis et al., 2024). Variation in reporting completeness that persists within a week acts on every case in that week and therefore behaves like environmental variation, not like sampling noise.

Sample size calculations for surveillance usually assume that more cases give proportionally more precision (Cochran, 1977), and analyses of reproduction number estimation have concentrated on the informativeness of incidence for period-specific values (Parag et al., 2022; Bhatt et al., 2023). The ecological partition of variance has, to our knowledge, not been carried over to the design of surveillance for the reproduction number, where it requires overdispersed offspring, binomial reporting with volatile completeness and a budget spread over several periods. An empirical test also faces a measurement problem. Weekly growth statistics computed from admissions are apparent quantities whose variance depends on the delay from infection to admission and on how each statistic aggregates growth, so their intercepts cannot be read directly as transmission variance. Observations from the same calendar week share national influences, and interval estimates that treat them as independent are too narrow (Cameron et al., 2011).

Here we extend the ecological variance partition to case-count estimates of the reproduction number under environmental variation in R_t, negative binomial offspring and binomial reporting, and derive the balance scale at which environmental and sampling contributions are equal. We examine its consequences for reporting heterogeneity and for dividing a fixed budget between cases per period and number of periods. We then estimate the scale in weekly influenza hospital admissions from NHSN with two apparent weekly statistics and map their intercepts to the latent environmental variance by a two-step inversion through a data-matched simulation. Finally, we decompose the intercept into national-week and state-specific components and compare the mandatory and voluntary reporting regimes.

2. Background

2.1. Transmission, reporting and two estimation targets

Consider transmission over one generation, indexed by t, in which n infected individuals act as primary cases. Conditional on the instantaneous reproduction number R_t, the number of secondary infections caused by primary case i, $X_{i,t}$, is assumed to follow a negative binomial distribution, which captures individual heterogeneity in transmission (Lloyd-Smith et al., 2005). The offspring distribution is given by:

$$X_{i,t} \mid R_t \sim \mathrm{NB}\left(k,\ \frac{k}{k+R_t}\right), \qquad \mathbb{E}(X_{i,t} \mid R_t) = R_t, \qquad \mathrm{Var}(X_{i,t} \mid R_t) = R_t + \frac{{R_t}^2}{k}, \tag{1}$$

where $k > 0$ is the dispersion parameter, smaller values of k correspond to greater heterogeneity, and $k \to \infty$ recovers Poisson offspring. Each infection is then reported independently with probability $\rho \in (0, 1]$. Because a sum of independent negative binomial variables with a common success probability is negative binomial, and because binomial thinning preserves this family, the reported count C_t satisfies:

$$C_t \mid n, R_t \sim \mathrm{NB}\left(nk,\ \frac{k}{k+\rho R_t}\right), \qquad \mathbb{E}(C_t \mid n, R_t) = \rho n R_t, \qquad \mathrm{Var}(C_t \mid n, R_t) = \rho n R_t + \frac{\rho^2 n {R_t}^2}{k}, \tag{2}$$

where $m = \rho n$ is the expected number of reported cases per period when $R_t = 1$. We refer to m as the case volume.

The model rests on four assumptions. Each infection is reported independently with a probability ρ that does not change over time (H1). Primary cases transmit independently and identically according to Eq. (1) (H2). The dispersion k does not depend on R_t or n (H3). The number of primary cases n is fixed by design and is independent of R_t (H4). Assumption H4 holds when n is set by contact tracing; in routine surveillance n is itself the outcome of earlier transmission, and the renewal formulation in Section 6.3 relaxes it.

Transmission potential varies between periods with weather, school terms, holidays and behaviour (Dalziel et al., 2018). Its marginal moments are denoted $\mathbb{E}(R_t) = \bar{R}$ and $\mathrm{Var}(R_t) = v_R$. Because this variation acts on every chain of transmission in the same period, it corresponds to environmental stochasticity in population ecology, whereas the offspring and reporting variation of Eq. (2) corresponds to demographic stochasticity (Engen et al., 1998; Lande et al., 2003).

The scaled count $\hat{R}_t = C_t/m$ can be used for two different targets. The first is the period-specific value R_t, which is the quantity of interest for situational awareness and for judging whether transmission is currently above one. The second is the mean across periods, $\bar{R}$, which is the quantity of interest when describing typical transmission in a season. When

R_t is independent across periods, $\bar{R}$ is also the best one-step-ahead predictor of next period's value. The error for the first target is $\hat{R}_t - R_t$. The error for the second target can be written as:

$$\hat{R}_t - \bar{R} = (\hat{R}_t - R_t) + (R_t - \bar{R}), \tag{3}$$

where the second term does not depend on m. Conditional on R_t, $\hat{R}_t$ is unbiased for R_t, and its variance $R_t/m + \rho {R_t}^2/(km)$ falls to zero as m grows; in this case $\hat{R}_t$ attains the Cramér–Rao bound for R_t (Appendix A).

2.2. Error variance and the balance scale

For the second target, the error variance follows from the law of total variance applied to Eq. (3).

Proposition 1 (Error variance for the mean across periods)**.** *Under H1–H4, the variance of $\hat{R}_t = C_t/(\rho n)$ about $\bar{R}$ is given by:*

$$V(n,\rho) = v_R + \frac{\bar{R}}{\rho n} + \frac{\bar{R}^2 + v_R}{kn} = v_R + \frac{A}{m}, \qquad A = \bar{R} + \frac{\rho(\bar{R}^2 + v_R)}{k}, \tag{4}$$

where the three terms are the environmental, reporting and offspring contributions. Then:

(i) for fixed ρ, $V \to v_R$ as $n \to \infty$;

(ii) for fixed n, $V \to v_R + \bar{R}/n + (\bar{R}^2 + v_R)/(kn) > v_R$ as $\rho \to 1$;

(iii) at the balance scale $m_\times = A/v_R$, the sampling and environmental contributions are equal, and doubling m reduces V by the fraction $1/[2(1 + m/m_\times)]$;

(iv) among estimators that are conditionally unbiased for R_t, no estimator has a smaller mean squared error for $\bar{R}$ than V.

The balance scale plays the role of the critical population size in ecological models, at which demographic and environmental contributions to the variance of population growth are equal (Lande et al., 2003). It is a reference point for comparing the two contributions and does not by itself define a minimum operating size. Part (iv) is restricted to conditionally unbiased estimators. An estimator that shrinks $\hat{R}_t$ towards a prior mean can have smaller error for $\bar{R}$, and averaging over T independent periods reduces the environmental term to v_R/T (Proposition 3).

Parts (i) and (ii) show that case volume can be raised in two ways with different limits. Coverage n removes both sampling terms in the limit, whereas full reporting leaves the offspring term. Near the baseline, a small proportional increase in n reduces V by $A/\bar{R}$ times as much as the same proportional increase in ρ. The limit v_R also assumes that environmental variation is independent between periods and between places. If R_t follows a stationary first-order autoregressive process with autocorrelation ϕ, the environmental part of one-step prediction error falls to $(1 - \phi^2)v_R$. If J jurisdictions share environmental variance v_R with mean pairwise correlation $\bar{c}$, the environmental variance of their average is given by:

$$v_R\,\bar{c} + \frac{v_R(1 - \bar{c})}{J}, \tag{5}$$

where pooling reduces this limit only to the extent that variation is not synchronised ($\bar{c} < 1$).

2.3. Reporting heterogeneity and the allocation of a surveillance budget

Reporting completeness is rarely constant. Assumption H1 can be replaced by a reporting probability p_t that varies between periods independently of R_t and n, with mean ρ and relative volatility $\tau = \mathrm{SD}(p_t)/\rho$, and with counts independent across periods conditional on R_t and p_t (H5).

Proposition 2 (Reporting heterogeneity)**.** *Under H2–H5, the error variance of $\hat{R}_t = C_t/m$ about $\bar{R}$ can be written as:*

$$V_H(m) = v_R + v_H + \frac{A_H}{m}, \qquad v_H = \tau^2(\bar{R}^2 + v_R), \qquad A_H = \bar{R} + \frac{\rho(1 + \tau^2)(\bar{R}^2 + v_R)}{k}, \tag{6}$$

where v_H is the variance added by reporting heterogeneity. The limiting variance rises to $v_R + v_H$, the balance scale falls to $m_{\times,H} = A_H/(v_R + v_H) < m_\times$, and v_R and v_H cannot be separated using the mean and variance of the counts alone, which identify only $(1 + \tau^2)(\bar{R}^2 + v_R)$.

Proposition 2 is the identification limit for the empirical part of this study. An intercept estimated from counts alone combines transmission and reporting variation unless reporting is measured separately.

The balance scale also determines how a fixed budget is best divided. Suppose that T periods are observed with case volume m in each, that each period costs c_0 and each reported case costs c_1, and that the budget is $\mathcal{B} = T(c_0 + c_1 m)$.

Proposition 3 (Allocation of a fixed budget)**.** *Under H1–H4 with R_t independent across periods, the variance of the mean of $\hat{R}_t$ over T periods is given by:*

$$\frac{v_R}{T} + \frac{A}{Tm}, \tag{7}$$

which, under the budget constraint, is minimised at:

$$m^* = \sqrt{\frac{c_0}{c_1}\, m_\times}, \tag{8}$$

where $T^* = \mathcal{B}/(c_0 + c_1 m^*)$, subject to $T^* \geq 1$. If R_t follows a stationary first-order autoregressive process with autocorrelation ϕ, the sampling term is unchanged and the environmental term becomes $v_R S(\phi, T)/T^2$ with

$$S(\phi, T) = \frac{T(1+\phi)}{1-\phi} - \frac{2\phi(1-\phi^T)}{(1-\phi)^2}. \tag{9}$$

The optimal case volume per period is the geometric mean of c_0/c_1 and the balance scale, the same form as the optimal number of secondary units in two-stage sampling (Cochran, 1977). Once m exceeds m^*, additional budget reduces error more when spent on further periods than on further cases. With autocorrelated environments the optimum also depends on the budget, and Appendix B gives the threshold at which it switches.

Reporting heterogeneity can reverse the ranking of coverage and completeness as targets for investment.

Corollary 1 (Coverage versus completeness)**.** *If τ does not depend on ρ, increasing coverage n reduces V_H more than the same proportional increase in ρ. If reporting variation arises from binomial sampling of ν units, so that* $\mathrm{Var}(p_t) = \rho(1-\rho)/(\nu+1)$*, increasing ρ is preferable whenever:*

$$m > \frac{\rho^2 \nu}{k}, \tag{10}$$

where ν is the number of reporting units sampled per period.

3. Results

3.1. Diminishing returns from case volume for the mean reproduction number

Under the baseline parameters ($\bar{R} = 1.15$, $v_R = 0.04$, $k = 0.35$, $\rho = 0.25$; see Materials and methods), the sampling coefficient was $A = 2.1232$ and the balance scale was $m_\times = 53.1$ reported cases per period (Fig. 1B). Error variance for the period-specific value fell in proportion to $1/m$ over the whole range, whereas error variance for the mean levelled off at v_R. Doubling the case volume reduced error variance for the mean by 42.1% at $m = 10$, by 25.0% at $m = m_\times$ and by 10.5% at $m = 200$. Monte Carlo estimates from 20000 replications per point differed from Eq. (4) by at most 4.0%.

Raising reporting probability at fixed coverage did not reduce error variance to v_R (Fig. 1C). At full reporting the residual variance was 0.166 with 40 primary cases, 0.0652 with 200 and 0.0450 with 1000, against $v_R = 0.04$. Near the baseline, a proportional increase in coverage reduced error variance 1.85 times as much as the same proportional increase in reporting probability. The balance scale was largest where environmental variance was small and offspring were strongly overdispersed (Fig. 1D).

A relative reporting volatility of $\tau = 0.15$ added $v_H = 0.0307$ to the limiting variance and reduced the balance scale from 53.1 to 30.4 cases (Fig. 2A, B). The scenario parameters for influenza admissions give latent balance scales of 21.3 to 49.2 cases per generation (see Materials and methods); Section 3.3 reports the latent value implied by the data. Over this range, the optimal case volume m^* of Eq. (8) was 14.6 to 22.2 cases per period for $c_0/c_1 = 10$ and 46.2 to 70.1 cases for $c_0/c_1 = 100$ (Fig. 2C). In a hypothetical example with ten periods of 50 cases, autocorrelated environment and reporting ($\phi = \psi = 0.5$) and $\tau = 0.15$, the mean squared error of the mean was 22.57×10^{-3}, of which 10.40×10^{-3} was environmental, 7.88×10^{-3} reporting and 4.29×10^{-3} sampling (Table S7). A tenfold increase in case volume reduced it to 18.71×10^{-3}, whereas standardising reporting reduced it to 14.65×10^{-3} and four times as many periods to 6.17×10^{-3}. At twice the baseline budget, spending the increment on cases gave 19.35×10^{-3} and spending it on periods gave 11.98×10^{-3} (Fig. 2D).

3.2. Calibration of apparent weekly statistics by simulation

The empirical statistics are apparent quantities. The weighted growth ratio G compounds more than one generation of growth into a week, and the renewal-type ratio R_A maps weekly growth back to the generation scale through an assumed generation interval (Section 6.2). Their intercepts therefore need a calibration before they can be compared with v_R. We built this calibration in a simulation that reproduces the observed panel (Section 6.8). It keeps the four seasons, the rule-window lengths and the initial admissions of each state, and adds a national week shock that carries 45% of v_R and the delay from infection to admission. We simulated 200 panels at each of four values of v_R. The mean fitted intercept rose

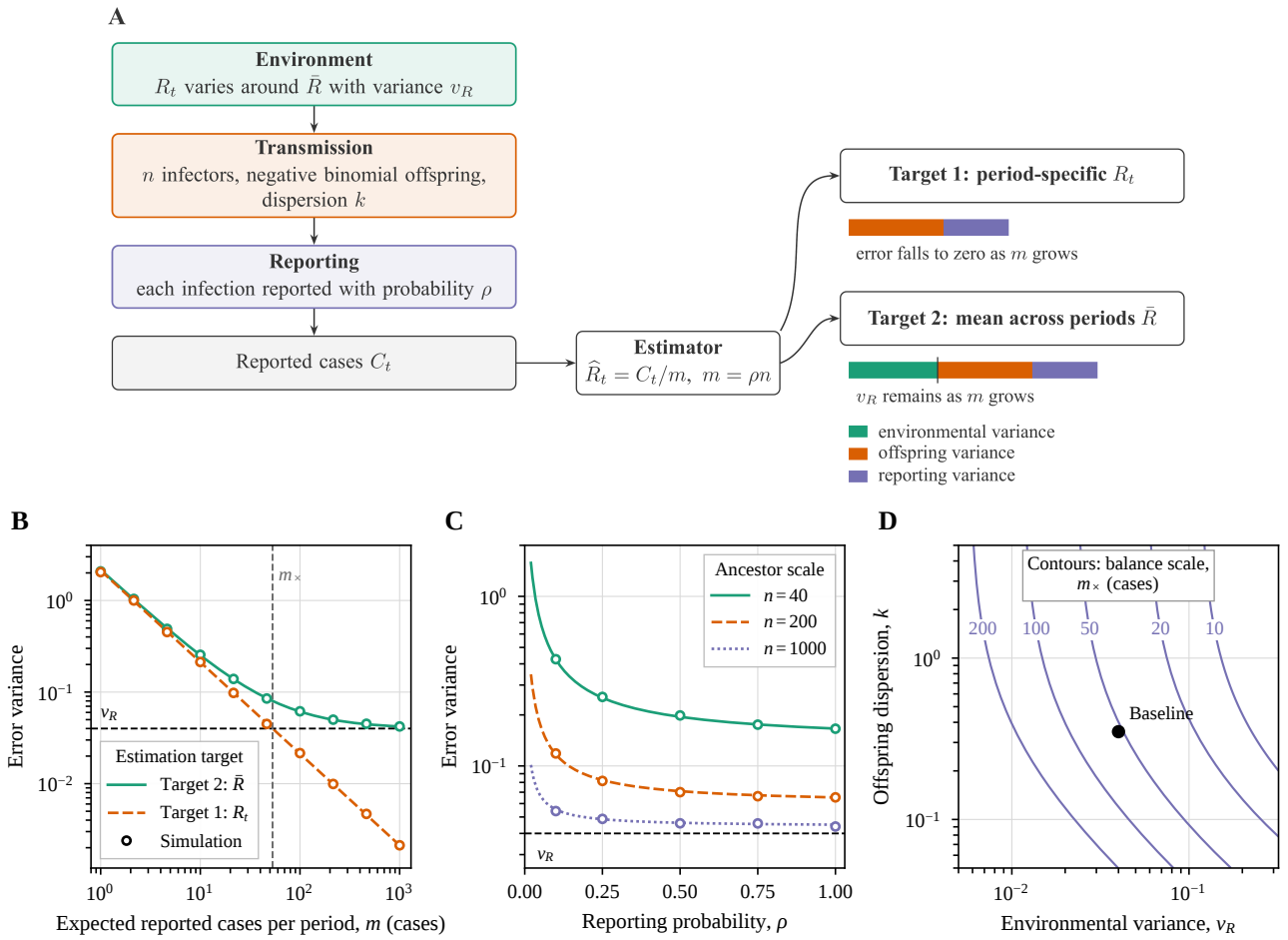


Fig. 1. Contributions of environmental, offspring and reporting variation to the error of the scaled estimator of the reproduction number. (A) Schematic of the two estimation targets, the period-specific value and the mean across periods, for the estimator that divides reported cases by expected case volume; colours mark the environmental, offspring and reporting sources of error. (B) Error variance against expected reported cases per period for the mean across periods and for the period-specific value; lines are closed forms, circles are Monte Carlo estimates with 95% intervals, and dashed lines mark the environmental variance v_R and the balance scale. (C) Error variance against reporting probability for three fixed numbers of primary cases; the value at full reporting is the residual offspring term. (D) Contours of the balance scale over environmental variance and offspring dispersion, with the baseline parameter set marked.

linearly with v_R for both statistics (Fig. 3A), and no cell mean departed from the fitted line by more than 0.60 Monte Carlo standard errors.

Expressed as a scale factor on $v_R/\bar{R}^2$, the fitted intercept was $\kappa = 0.516$ for R_A and 2.729 for G, close to the noise-free limits of 0.493 and 2.745 (Fig. 3B). A simpler design with 500 independent states and six observation weeks gave noise-free limits of 0.485 and 2.698, in line with the medians of 0.484 and 2.68 over the 17 scenarios of the first simulation. The data-matched panels also reproduced the dependence structure of the data. The intraclass correlation of weekly losses within calendar weeks was 0.258 for R_A and 0.310 for G in simulation, against 0.253 and 0.310 in the data, and the national-week share of the simulated R_A intercept was 0.43, against 0.41 observed. Across settings that changed one ingredient at a time, κ for R_A moved most with the true generation interval (1.287 for a mean of 2.2 days and 0.380 for 4.3 days, with the statistic still computed at 3.6 days), with autocorrelation of transmission between weeks (0.780 at $\phi = 0.5$) and with a synchronous shock to reported counts (1.041). It changed little with the national share of the shock, the admission delay or the offspring dispersion (0.454 to 0.535; Table S8).

3.3. Empirical intercepts and the implied environmental variance

In the rule-defined early-growth windows of four seasons, 1748 state-weeks in 42 calendar weeks gave a renewal-type ratio intercept of $a = 0.0150$ (95% CI: 0.0100, 0.0208), slope $b = 0.560$ (95% CI: 0.367, 0.769) and $b/a = 37.4$ (95% CI: 19.1, 71.0) (Fig. 4B; Table S1). The weighted growth ratio in the same windows gave $a = 0.0840$ (95% CI: 0.0498, 0.1248) and $b/a = 23.5$ (95% CI: 8.4, 59.7) from 1756 state-weeks. The eight additional state-weeks have a reporting scale D_t of at least five but fewer than five admissions in the current week, which R_A requires because its scale is the single-week count. In the three calendar windows of the earlier analysis, the growth ratio gave $a = 0.0991$ (95% CI: 0.0569, 0.1382) and $b/a = 22.6$ (95% CI: 9.0, 52.2). All bootstrap replicates gave a positive intercept for both rule-window specifications, as did 99.95% for the calendar windows. The bootstrap standard deviation of a was 1.95 to 4.68

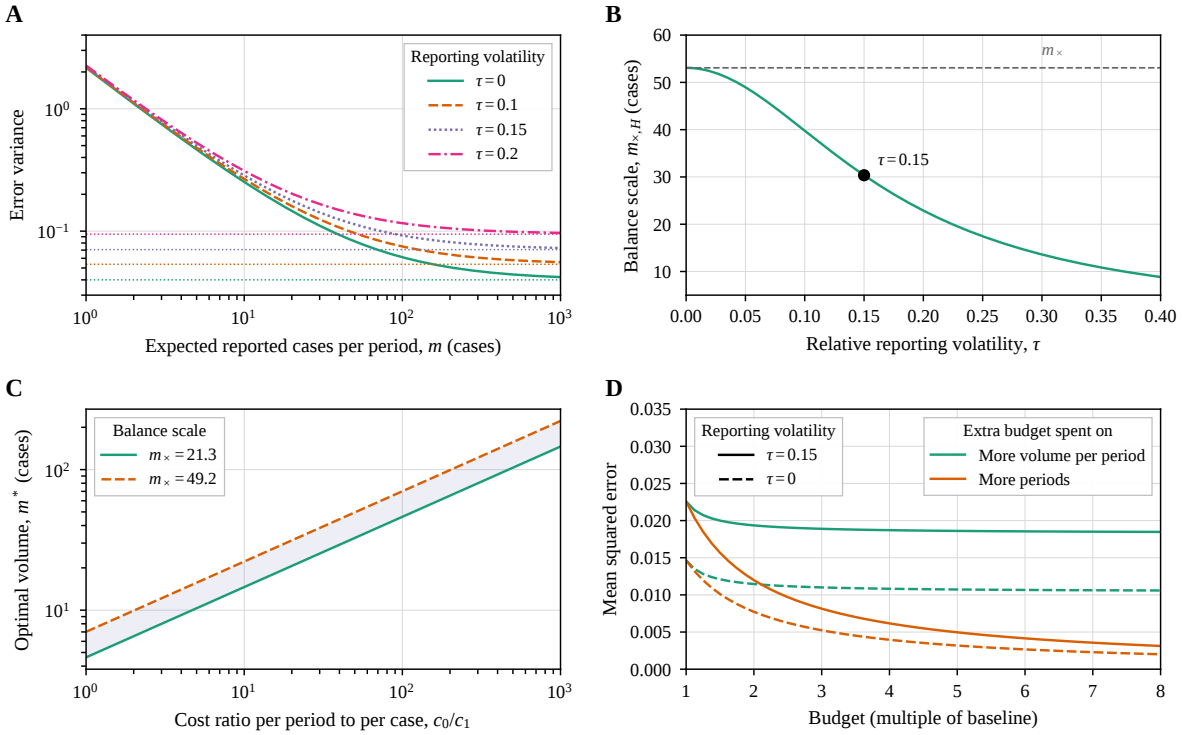


Fig. 2. Reporting heterogeneity and the allocation of a fixed surveillance budget. (A) Error variance against expected reported cases per period for four levels of relative reporting volatility; dotted lines mark the corresponding limits as case volume grows, set by environmental plus reporting variance. (B) Generalised balance scale against relative reporting volatility; the dashed line is the balance scale without reporting volatility, and the dot marks the volatility used in panel D. (C) Optimal number of reported cases per period against the price ratio c_0/c_1, for the two ends of the scenario range of the balance scale; the band spans the range between them. (D) Mean squared error of the estimator of the mean reproduction number against total budget, when additional budget buys either more cases per period or more periods, with autocorrelated environmental and reporting variation, under baseline (solid) and standardised (dashed) reporting.

times that obtained by resampling states only.

The ratio b/a varied across specifications by a factor of 3.9 for R_A, from 24.7 after trimming the most influential 1% of losses to 97.1 in peak weeks, and by a factor of 7.1 for G, from 12.2 with equal denominator weights to 86.8 in peak weeks (Fig. 4D). Generation interval variants changed it by less than 3% for R_A. Over the observed range of scales, a power law without an intercept fitted the binned losses about as well as the affine form (bootstrap $p = 0.114$ for R_A, 0.438 and 0.418 for G; Table S2). The affine form was retained because its parameters correspond to Proposition 1 and enter the calibration directly.

Mapping each empirical intercept through its calibration line gave an implied latent variance of $v_R^{\text{impl}} = 0.040$ (95% CI: 0.026, 0.056) for R_A and 0.043 (95% CI: 0.025, 0.064) for G (Fig. 3A, C). Bootstrap variation in the empirical intercept accounted for nearly all of this width, since the interval from calibration error alone was 0.0397 to 0.0406 for R_A. The scale factors of the two statistics differ by a factor of 5.3. Their implied variances differed by -0.003 (95% CI: -0.012, 0.005), a ratio of 0.93 (95% CI: 0.78, 1.18). The 90% interval of this ratio, 0.81 to 1.13, lies within 0.80 to 1.25, so two one-sided tests at the 5% level declare the two estimates equivalent within a margin of 20%. Both point values lie at the lower edge of the band of 0.04 to 0.08 used in the scenario calculations, and both intervals extend below it. On the latent scale, Proposition 1 at the simulator values ($\bar{R} = 1.18$, $\rho = \rho_h = 0.01$, $k = 0.35$) gives a balance scale of 30.3 admissions per generation (95% CI: 21.7, 46.3) for R_A, or about 59 per week at 1.94 generations per week. The apparent ratio of 37.4 is defined on the scale of R_A and does not estimate this latent quantity directly.

Across ten settings, v_R^{impl} ranged from 0.013 to 0.063 (Fig. 3C; Table S8). The true generation interval had the largest effect, giving 0.013 and 0.014 when it was shorter than the 3.6 days assumed by the statistic and 0.058 and 0.063 when it was longer. The 95% interval of the difference between the two statistics contained zero in nine of the ten settings; this non-rejection is the sense of agreement used in Table S8. The exception replaced the transmission shock by a synchronous multiplicative shock to reported counts, which gave 0.019 for R_A and 0.029 for G (ratio 0.67; 95% CI: 0.39, 0.90).

Three further analyses examined the assumptions of the inversion (Table S10). The lag-1 autocorrelation of deviations from state-season means was -0.095 (95% CI: -0.168, -0.019) for R_A and 0.123 (95% CI: 0.039, 0.201) for G. Both statistics are autocorrelated by construction even when weekly shocks are independent; simulated panels with independent shocks gave means of -0.074 and 0.114, which placed the data at the 31st and 62nd percentiles. With a first-order coefficient of 0.25 the data fell at the 3rd and 10th percentiles, and with 0.5 below all simulated panels. The serial interval of seasonal influenza in households has been estimated at 3.6 days (95% CI: 2.9, 4.3) (Cowling et al., 2009). With a true mean generation interval of 2.9 days, the implied variance was 0.026 (95% CI: 0.017, 0.036) for R_A and 0.027 (95%

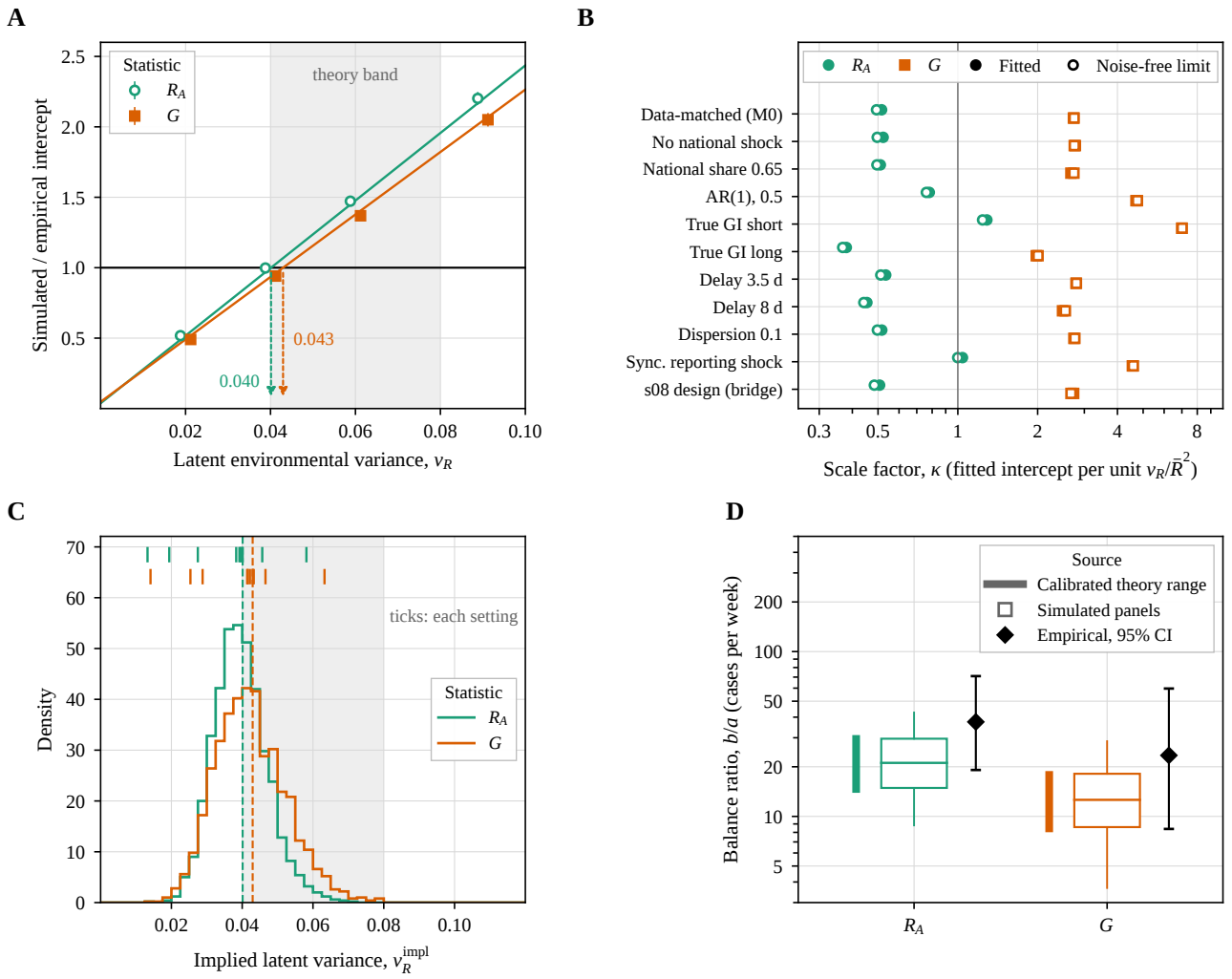


Fig. 3. Simulation-based inversion from apparent statistics to latent dynamics. (A) Calibration lines from 200 simulated panels per cell that reproduce the observed seasons, window lengths, state sizes, national week shocks and reporting delay. Fitted intercepts are scaled by the empirical intercept. Dashed drop lines mark where each line meets the data, giving the implied latent variance v_R. (B) Scale factor κ linking the fitted intercept to $v_R/\bar{R}^2$ for each simulation setting. Filled markers are fitted values and open markers the noise-free limit. (C) Distribution of the implied v_R under the data-matched setting. Uncertainty combines the two-way bootstrap of the data with Monte Carlo error of the calibration. Top ticks show the point values of the other settings. (D) Simulated per-panel b/a and the calibrated theoretical range, against empirical estimates with 95% intervals. Grey bands mark $v_R = 0.04$–0.08.

CI: 0.016, 0.040) for G; with 4.3 days it was 0.058 and 0.063. The autocorrelation of the data sat at the 2.5th and 9th percentiles of panels simulated with 2.9 days and at the 80th and 92nd percentiles with 4.3 days, so it does not favour the shorter interval. The shorter interval did reproduce the empirical b/a: at $v_R = 0.02$, the simulated value nearest its implied variance, the data sat at the 52.5th and 58th percentiles of the simulated b/a. Overdispersed weekly admissions with a variance-to-mean ratio of 1.5 or 2.0 kept the implied variance between 0.039 and 0.045. They raised the simulated b/a at $v_R = 0.04$ from 28.0 to 68.0 and 81.0 for R_A, and from 16.2 to 40.2 and 48.4 for G. These settings also gave lag-1 autocorrelations below the data for both statistics, so the data favour overdispersion smaller than a variance-to-mean ratio of 1.5.

On the scale of each statistic, the calibrated theory gave b/a of 13.9 to 31.1 for R_A and 8.0 to 18.8 for G over eight combinations of $v_R \in \{0.04, 0.08\}$, $k \in \{0.2, 0.5\}$ and admission probability $\rho_h \in \{0.0056, 0.021\}$ (Fig. 3D). The empirical point estimates of 37.4 and 23.5 lie above these ranges, and their 95% intervals overlap them. Ratios from single simulated panels varied widely; 95% of them lay between 8.8 and 42.7 for R_A and between 3.6 and 28.6 for G, and the empirical values fall at the 93rd and 91st percentiles of these distributions.

3.4. Multiscale decomposition of surveillance uncertainty

The intercept contains a component shared by all states in a calendar week and a component specific to states (Fig. 4C; Table S2). Removing national-week effects reduced a by 41% (95% CI: 12, 59) for R_A and by 51% (95% CI: 21, 67) for G in rule windows, and by 52% for G in calendar windows. Removing state-season effects as well changed the intercept little; for R_A it was 0.0083 (95% CI: 0.0064, 0.0141) after both adjustments, and the interval for the state-season component included zero in every specification. The intraclass correlation of the loss within calendar weeks was 0.253 for

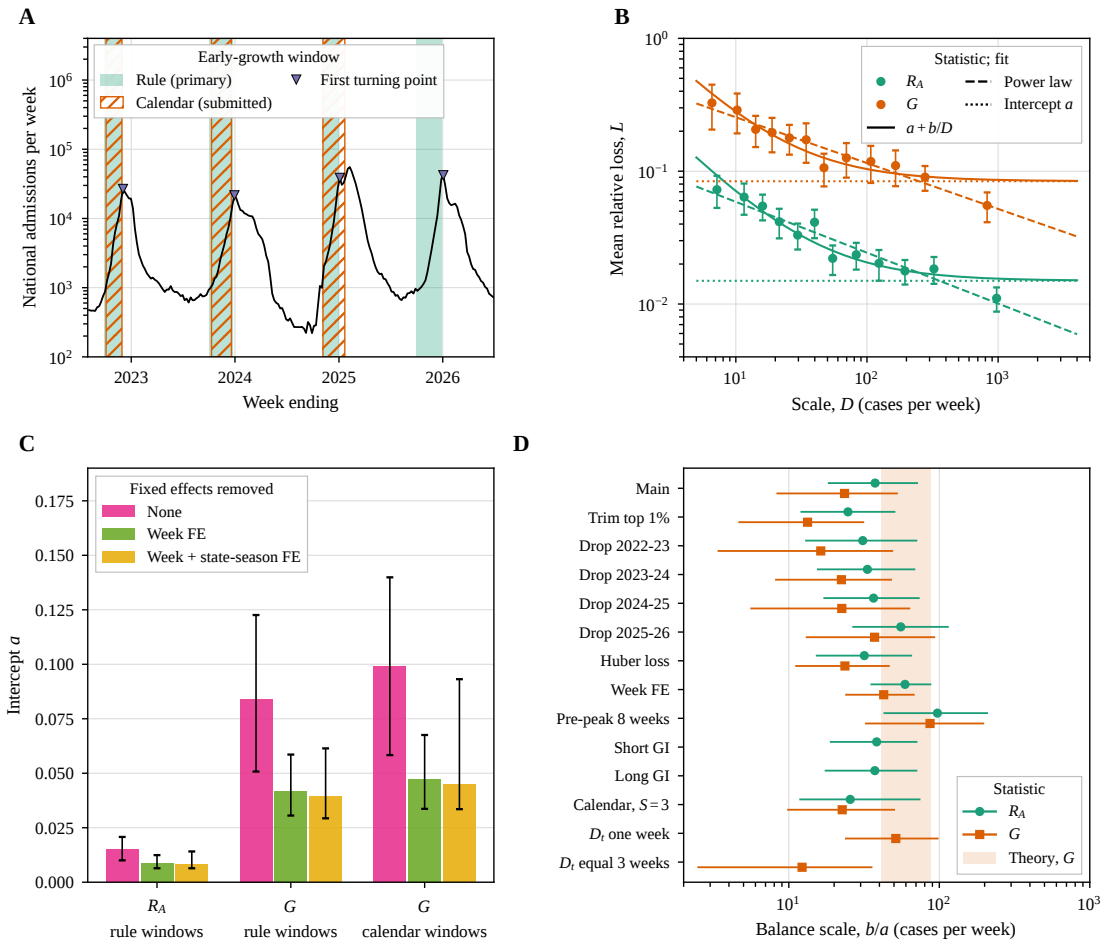


Fig. 4. Relative loss of two apparent weekly statistics in influenza hospital admissions. (A) National weekly admissions on a logarithmic scale; shading marks the rule-based early-growth windows of the primary analysis, hatching marks the calendar windows of the earlier analysis, and triangles mark the first national turning point of each season. (B) Binned mean relative loss against reporting scale for the renewal-type ratio and the weighted growth ratio, with the fitted affine curve, a power law without intercept used for comparison and the fitted intercept; bin error bars ignore clustering. (C) Intercept before adjustment, after removing national-week effects and after also removing state-season effects, with two-way bootstrap intervals. (D) Ratio of slope to intercept under sensitivity variants; the band is the uncalibrated theoretical range for the growth ratio with the denominator correction, and Fig. 3D gives the calibrated range.

R_A and 0.310 for G. The bootstrap reflected the same structure. Resampling weeks alone gave a standard deviation of $\hat{a}$ of 0.0021 for R_A and 0.0158 for G, 1.46 and 1.87 times the values from resampling states alone, and two-way resampling gave 0.0028 and 0.0193.

The national-week component enters the loss of every state in the same week. More admissions in one state reduce only that state's count term, and averaging over states reduces only the state-specific term, so neither removes the shared component; only further weeks average it out. Pooling the mean weekly deviation over J randomly chosen states reduced its variance to 0.39, 0.31 and 0.25 of the single-state value for R_A at $J = 5$, 10 and 51, and to 0.43, 0.36 and 0.31 for G (Fig. 6C). These values approach the national-week component, which is 0.23 of single-state variance for R_A and 0.29 for G, and come within 10% of it at $J = 33$ and $J = 25$ respectively, as Eq. (5) predicts for partly synchronised variation.

3.5. Mandatory and voluntary reporting regimes

Among state-weeks with a reporting scale of at least five, the median share of hospitals reporting was 92.3% in the first mandatory period, 45.7% under voluntary reporting and 92.9% after mandatory reporting resumed (Fig. 5A). In the rule windows the median was 93.7%, and 79.8% of state-weeks had at least 90% of hospitals reporting. Within these windows, adjusting each statistic for the reporting share changed the intercept of R_A by $\Delta a = 0.29 \times 10^{-3}$ (90% CI: -0.22×10^{-3}, 0.85×10^{-3}) and that of G by 0.37×10^{-3} (90% CI: -2.83×10^{-3}, 2.81×10^{-3}) (Fig. 5B; Table S3). Both changes were equivalent to zero within margins set at a relative reporting volatility of $\tau_0 = 0.10$. These margins equal 17% (95% CI: 12, 25) of the intercept of R_A and 12% (95% CI: 8, 20) of that of G, and they depend on the sample mean of each statistic through Eq. (16). The measured change was 2.0% of the intercept for R_A (95% CI: -2.4, 6.1; one-sided 95% upper bound 5.3%) and 0.4% for G (95% CI: -4.4, 4.0; upper bound 3.5%), and all four seasons for R_A and three of four for G were equivalent individually. At $\tau_0 = 0.05$, with margins of about 4% and 3% of the intercepts, neither change was equivalent, and the power of the test at a true change of zero was 0.25 for R_A and 0 for G (Fig. 5C).

The voluntary period gave the only contrast in which reporting completeness was low. Over May to October, the

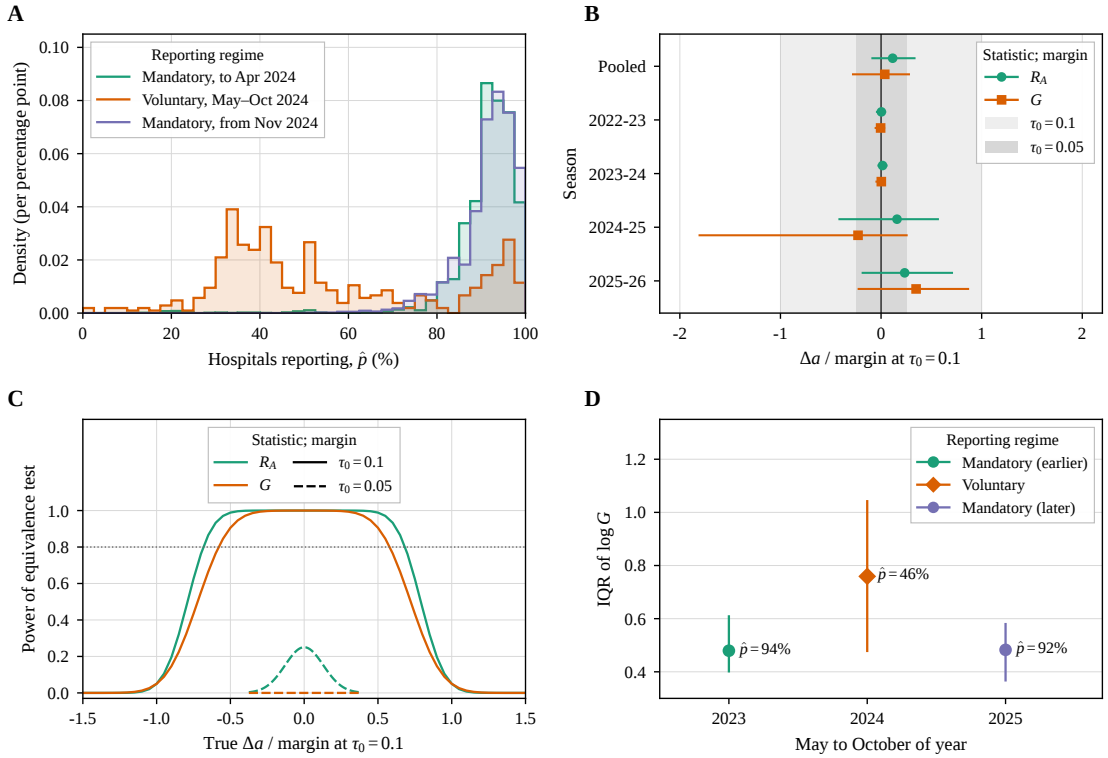


Fig. 5. Reporting completeness and the empirical intercept. (A) Distribution of the share of hospitals reporting influenza admissions under the three reporting regimes. (B) Change in the intercept when the statistic is adjusted for reporting coverage, pooled and by season, with 90% paired bootstrap intervals; each statistic is expressed relative to its own equivalence margin, and shaded bands show the margins implied by two thresholds of reporting volatility. (C) Power of the two one-sided equivalence test against the true change, for both statistics and both margins, with a conventional power reference. (D) Interquartile range of the logarithm of the growth ratio over May to October in three calendar years; the year with voluntary reporting is highlighted, and labels give median reporting completeness.

interquartile range of log G was 0.759 in 2024 under voluntary reporting, against 0.479 in 2023 and 0.483 in 2025 (Fig. 5D; Table S4). The ratio of 2024 to 2023 was 1.58 (95% CI: 0.95, 2.23) and that of 2024 to 2025 was 1.57 (95% CI: 0.95, 2.39), whereas the placebo ratio of 2025 to 2023 was 1.01 (95% CI: 0.72, 1.25). Median reporting scales in these months were similar (15.6, 16.6 and 16.2 admissions), and matching state-weeks on reporting scale gave ratios of 1.68 and 1.58. Matching on calendar month gave a median ratio of 1.33, with the ratio above one in nine of twelve month-by-year comparisons. COVID-19 admissions showed ratios of 1.62 against 2023 and 1.18 against 2025, and RSV admissions a ratio of 1.36 against 2025. In 44 states with laboratory data, laboratory R_A correlated with hospital R_A at $r = 0.31$ and shared 0.230 (95% CI: 0.043, 0.418) of the hospital intercept.

3.6. Diminishing returns and surveillance design

Under $V(m) = a + b/m$, the share of the precision attainable at unlimited case volume is $m/(m + m_\times)$, and one doubling of m reduces variance by the fraction $m_\times/[2(m + m_\times)]$ (Proposition 1). The share is 33%, 50%, 67% and 80% at $m = 0.5$, 1, 2 and 4 times $m_\times$, a doubling from these points cuts variance by 33%, 25%, 17% and 10%, and 90% of attainable precision requires $m = 9m_\times$ (Fig. 6B). With the empirical $m_\times$ of 37.4 for R_A and 23.5 for G, 55% and 65% of state-weeks in the rule windows lay above the balance scale. For a single state, the number of observation weeks needed to reach a given relative standard error of mean R_A fell steeply with admissions below $m_\times$ and changed little above about $4m_\times$ (Fig. 6A). At the empirical balance scale and a cost ratio of $c_0/c_1 = 10$, the optimal case volume was $m^* = 19.4$ (95% CI: 13.8, 26.6) for R_A and 15.3 for G; at $c_0/c_1 = 100$ it was 61.2 and 48.4.

The fitted variance departs from that of a Poisson renewal model, in which the error variance of the estimate is $\bar{R}/m$ and has no term that persists as m grows (Fig. 6D). On the apparent scale of R_A, the fitted slope was 1.25 times the Poisson value of 0.446, and the fitted variance was twice the Poisson variance at $m = 22$ and ten times at $m = 261$. On the latent scale of Proposition 1, with $v_R = 0.040$, the corresponding values were $m = 28$ and $m = 262$.

Seasons limit any estimate of variation between seasons (Fig. 7). Season-specific intercepts of R_A were 0.0137, 0.0062, 0.0127 and 0.0225, with a random-effects mean of 0.0125 (95% CI: 0.0022, 0.0227) and $I^2 = 0.47$ (Table S5). For G the random-effects mean was 0.0619 (95% CI: 0.0066, 0.1171) in rule windows and 0.0735 (95% CI: −0.0558, 0.2028) in calendar windows. The relative standard error of any unbiased estimate of the between-season variance is at least 1.00 with three seasons and 0.82 with four (Corollary 2). Intercept ratios for alternative denominator weights showed $q_1 < 1 < q_2$ for influenza, COVID-19 and RSV, for example $q_1 = 0.688$ and $q_2 = 1.539$ for influenza in calendar windows (Table S6). Epidemic growth with reporting noise and no environmental variation gave ranges of 0.63

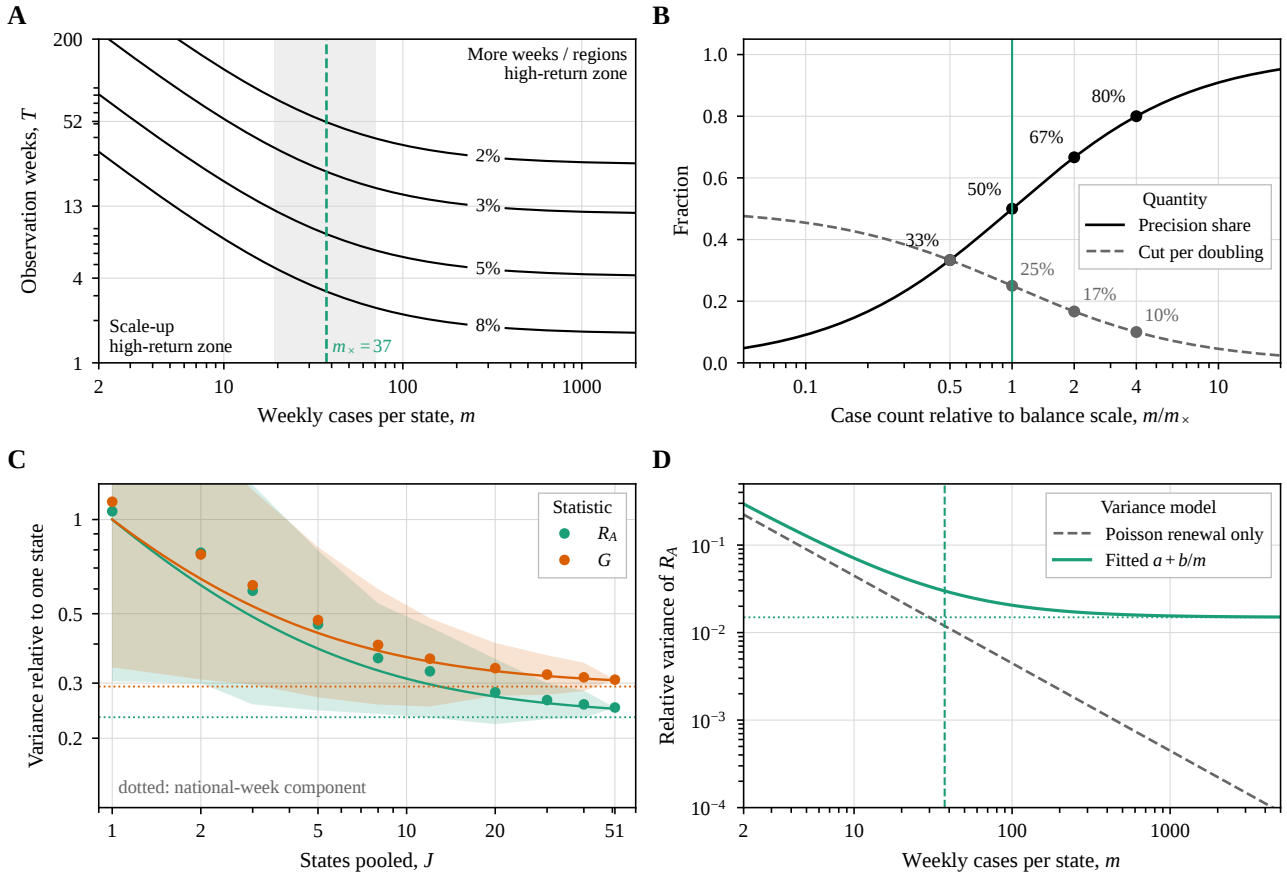


Fig. 6. Surveillance design nomogram. (A) Observation weeks needed for the relative standard error of a single state's mean renewal ratio R_A to reach 2–8%, by weekly admissions m, on the scale of the statistic and ignoring autocorrelation within states. The dashed line marks the balance scale $m_\times = b/a$, and the grey band its 95% interval. To its left, added cases give most of the gain. To its right, contours flatten and added weeks or regions give most of the gain. (B) Share of attainable precision, $m/(m + m_\times)$, and the variance reduction from one doubling of m. (C) Variance of the mean weekly deviation pooled over J states, relative to one state, for R_A and the growth ratio G. Points and bands come from random state subsets; dotted lines show the national-week component. (D) Relative variance of R_A: Poisson renewal variance alone against the fitted $a + b/m$.

to 0.89 for q_1 and 1.13 to 1.58 for q_2 in calendar windows, so this ordering does not discriminate between mechanisms. Of 40 one-sided tests, 21 were significant before correction, 18 after Benjamini–Hochberg correction and 10 after Holm correction, all of them for influenza and none for departures from the null ranges.

4. Discussion

We have described how the error of a case-count estimate of the reproduction number depends on case volume when transmission varies between periods, offspring numbers are overdispersed and only a proportion of infections is reported. For the period-specific value, error falls without limit as reported cases accumulate. For the mean across periods and for one-step prediction, error per period approaches the environmental variance, and the balance scale $m_\times = A/v_R$ marks the case volume beyond which each doubling of cases removes at most a quarter of the remaining variance. In weekly influenza hospital admissions over four seasons, two apparent statistics had positive intercepts, and a two-step inversion through a data-matched simulation mapped both to a latent environmental variance of about 0.04. A national-week component accounted for about 40% to 50% of the intercept, and reporting completeness under mandatory reporting contributed less than an equivalence margin of 17% of the intercept, set at a reference reporting volatility of 0.10.

The theory extends a standard result in population ecology to surveillance data. Ecological models split the variance of population growth into a demographic term that falls with population size and an environmental term that does not (Engen et al., 1998; Lande et al., 2003). The present model retains this partition and adds four elements that matter for surveillance. Transmission is a branching process with negative binomial offspring, so that superspreading enlarges the demographic term (Lloyd-Smith et al., 2005). Binomial reporting adds a further term that falls with case volume, and reporting volatility adds one that does not (Proposition 2). The quantity of interest is an estimate averaged over periods, so the balance scale also sets the optimal division of a budget between cases per period and number of periods (Proposition 3). That rule has the form of optimal allocation in two-stage sampling (Cochran, 1977), with the balance scale in place of the ratio of within-unit to between-unit variance.

The results complement the information-theoretic analyses of Parag and colleagues (Parag, 2021; Parag et al., 2022).

Their metric for a noisy epidemic curve already contains time-varying reporting fractions, which enter through the geometric mean of the reporting probabilities, and a negative binomial extension is treated in their supplementary material (Parag et al., 2022). Their target is the sequence of period-specific values, and in a Poisson renewal model the error variance of each estimate is $\bar{R}/m$, so that reporting acts by rescaling the effective case count. Here the target is the mean across periods, and variation shared by all chains of transmission in a period adds a term that does not fall with m. The two descriptions agree at small counts and separate near the balance scale. On the apparent scale of R_A, the fitted variance was twice the Poisson variance at 22 admissions per week and ten times at 261, and on the latent scale the corresponding values were 28 and 262. Both descriptions are therefore needed: the Poisson analysis ranks data streams by the information they carry about current transmission, and the variance partition here indicates when further cases cease to improve estimates of typical transmission.

The two-step inversion connects the apparent statistics with the latent model. The simulation first established that neither statistic estimates $v_R/\bar{R}^2$ directly; the intercept of G was 2.7 times this quantity and that of R_A about half of it, for reasons that follow from how each statistic aggregates growth over a week. After calibration, the two statistics implied latent variances of 0.043 and 0.040 (ratio 0.93; 95% CI: 0.78, 1.18). This consistency is expected if both statistics respond to the same latent variation and the simulation captures how each transforms it. As a test of the calibration it has limited power: across the simulated settings it failed only when a synchronous shock to reported counts replaced the shock to transmission, and in that setting the observed lag-1 autocorrelation of G lay above 99% of simulated panels. The simulated panels further reproduced the intraclass correlation, the national-week share and the lag-1 autocorrelation of the data. Within the stated uncertainties, the calibration is therefore consistent with the empirical intercepts, under three conditions. The implied variance depends on the true generation interval. Across the 95% interval of 2.9 to 4.3 days for the serial interval of influenza (Cowling et al., 2009), it ranged from 0.026 to 0.063, so the inversion identifies v_R conditionally on that interval. Weekly shocks were taken as independent. The observed autocorrelation lay near the centre of the simulated distribution under independence, and below 97% of simulated panels with a first-order coefficient of 0.25 for R_A; a coefficient of 0.5 would have lowered the implied variance to about 0.026. Finally, the calibrated theoretical b/a lies below both empirical point estimates, which fall at the 91st and 93rd percentiles of the simulated sampling distribution. Overdispersion in weekly admissions can account for this excess. With a variance-to-mean ratio of 1.5, the simulated b/a rose to 68.0 for R_A and 40.2 for G, above both empirical values, while the implied variance stayed at 0.040 and 0.044. A true generation interval of 2.9 days also reproduced the empirical b/a, with an implied variance of 0.026. The excess can therefore come from overdispersion smaller than a variance-to-mean ratio of 1.5 or from a shorter generation interval. The two explanations imply variances of about 0.040 and 0.026. The autocorrelation of the data argues against the shorter interval but does not exclude it. The data cannot exclude a mixture of transmission and reporting shocks (Proposition 2).

The empirical results are best read as a diminishing-returns scale for case volume within a period. With an apparent balance scale of 37.4 admissions per week on the scale of R_A, a state that already reports at the balance scale obtains half of the precision that unlimited case volume could provide, and 90% requires nine times as many admissions. More than half of the state-weeks in the analysis windows were already above this scale. The limit applies to case volume per period. The error of an estimate pooled over T independent periods falls as v_R/T, estimators that shrink towards a prior mean can go below the per-period value (Appendix B.5), and a power law without an intercept described the observed range of scales about as well as the affine form. The balance scale is therefore a statement about where returns diminish, and it does not imply that error stops falling with more data of every kind. This interpretation is also robust to the source of the intercept. Whatever mixture of transmission, testing and statistic construction it contains, a component that is shared within a period and does not fall with case volume limits the value of further cases per period and raises the value of further periods.

The multiscale decomposition locates this component. About 41% of the intercept of R_A and 51% of that of G was common to all states in a calendar week, and resampling weeks moved the estimated intercept 1.5 to 1.9 times as much as resampling states. A national-week component cannot be reduced by scaling up surveillance in a single state, because it enters every state's loss in that week. Pooling states removes the state-specific part, and in these data pooled variance approached the national-week component, about a quarter of single-state variance, with 25 to 33 states. This share is consistent with national influences such as weather, school calendars and holidays, and with national changes in testing or admission practice (Dalziel et al., 2018). Two-way inference is required for the same reason; intervals that resample states only were about two to five times too narrow for the intercept (Cameron et al., 2011).

Within the limits of an observational comparison, the contrast between reporting regimes is consistent with a benefit of high facility participation. Under mandatory reporting, a median of about 93% of hospitals reported. Adjustment for reporting completeness changed the intercept of R_A by 2%, with an upper bound of 5%, and the change was equivalent to zero within a margin of 17%. During the voluntary period of 2024, when the median share fell to about 46%, the interquartile range of log growth was 1.58 times that of the same months in 2023. The interval for this ratio, 0.95 to 2.23, includes one, and the comparison rests on a single off-season. Several checks address the main confounders. The placebo comparison of 2025 with 2023 gave 1.01, matching on reporting scale and on calendar month preserved a ratio above one, and COVID-19 and RSV admissions showed ratios above one in the same period. These results are consistent with an association between low and variable facility participation and more volatile growth statistics, and they do not identify a causal effect of the reporting mandate. The adjustment measures facility participation only. Testing intensity, coding

and care seeking within reporting hospitals are not captured. The laboratory proxy shared 23% of the intercept of R_A, but it does not discriminate between transmission and ascertainment, because laboratory growth carries its own noise and attenuates the shared fraction towards zero.

These results suggest a design rule for routine surveillance of the mean reproduction number. Where weekly admissions per jurisdiction already exceed the balance scale, additional resources reduce error more when spent on further periods and on further jurisdictions whose variation is not synchronised (Eq. (5)) than on further cases per period. The value of complete reporting depends on the reporting volatility a system actually has. In the hypothetical example of Section 3.1, with a relative volatility of $\tau = 0.15$, standardising reporting reduced error more than a tenfold increase in case volume per period. The completeness bound observed under mandatory reporting allows τ up to about 0.04, and at that value a tenfold increase in cases reduced the mean squared error from 15.2×10^{-3} to 11.4×10^{-3}, against 14.6×10^{-3} for standardised reporting. In these data, maintaining high facility participation is therefore associated with lower volatility, and its value relative to case volume grows with the volatility that incomplete participation introduces. Seasons are the limiting unit for statements about variation between seasons. With four seasons, no unbiased estimator of a between-season variance can have a relative standard error below 0.82 (Corollary 2), and the intercept ratios for alternative denominator weights did not discriminate between mechanisms, because epidemic growth with reporting noise alone already produced $q_1 < 1 < q_2$.

The scope of these conclusions is set by the data and the calibration. The empirical results rest on one surveillance system, four influenza seasons and rule-defined early-growth windows. The window rule was revised once and the renewal-type ratio became the primary statistic during the analysis; Table S9 lists each specification with its role and date, and the headline estimates lie within each other's intervals. The inversion is conditional on the generation interval and on the simulated delay and offspring processes, and R_A is a transformation of weekly growth, not a posterior estimate of R_t (Cori et al., 2013). The propositions assume environmental variation that is independent between periods and a fixed number of primary cases; the renewal formulation of Section 6.3 shows that uncertainty in infection pressure adds a further term that does not fall with case volume. The equivalence margins use the sample mean of each statistic, the null ranges for the intercept ratios in rule windows come from the calendar-window simulation, and the bin error bars in Fig. 4B ignore clustering. Three extensions would sharpen these conclusions: renewal estimates with an explicit admission delay and an externally estimated generation interval, hierarchical models with explicit week effects combined with independent measures of testing, and a second surveillance system with mandatory reporting and more seasons.

5. Conclusion

When the target of surveillance is typical transmission or next period's value, the precision of a case-count estimate of the reproduction number grows with case volume at a diminishing rate. Beyond the balance scale, variation shared within a period dominates and each further doubling of cases removes at most a quarter of the remaining variance. In US influenza hospital admissions this scale was about 37 admissions per state-week on the scale of a renewal-type ratio. Conditional on a 3.6-day generation interval and on independent weekly shocks, the implied environmental variance was about 0.04 after calibration by simulation, and a large part of the shared variation was common to all states in a week. Surveillance systems that already report more cases per period than the balance scale gain more precision from more periods and from more regions with independent variation than from more cases per period. The evidence on reporting is narrower: under mandatory reporting, completeness contributed little to the intercept, and in an observational comparison high facility participation was associated with less volatile growth statistics. Empirical intercepts of apparent statistics should be reported together with the calibration that links them to transmission.

6. Materials and methods

6.1. Data and analysis windows

Weekly laboratory-confirmed influenza hospital admissions and the percentage of hospitals reporting them were taken from the NHSN Weekly Hospital Respiratory Data table for the 50 US states and the District of Columbia (Centers for Disease Control and Prevention, 2026), using the version retrieved on 2026-10-20. The table contained 21641 jurisdiction-weeks from 2020-08-08 to 2026-10-10. Reporting was mandatory until 30 April 2024, voluntary from 1 May to 31 October 2024 and mandatory again from 1 November 2024, giving 6009, 420 and 4077 jurisdiction-weeks with a reporting scale of at least five in the three regimes.

The primary windows covered early epidemic growth in four seasons from 2022–23 to 2025–26. Each window contained the contiguous weeks from 1 October, or from 9 November 2024 when mandatory reporting resumed, in which national weekly growth was at least 1.05, and ended the week before the first national turning point. The windows ran from 2022-10-01 to 2022-11-26, 2023-10-07 to 2023-12-23, 2024-11-09 to 2024-12-28 and 2025-10-04 to 2025-12-27, giving 42 calendar weeks. This rule replaced a first version that ended each window at the season maximum and gave only four weeks in the double-peaked 2024–25 season; the revision was made on 2026-10-20 before any intercept was fitted on the new windows. For continuity, the three calendar windows of an earlier version of this analysis (2022-10-08 to 2022-11-26, 2023-10-14 to 2023-12-16 and 2024-11-09 to 2025-01-18) were analysed as a secondary specification.

A peak window contained the national peak week of each season and the seven weeks after it. No zero-imputed counts entered any window.

6.2. Weekly growth statistics

Two statistics of weekly growth were computed for each state and week. The weighted growth ratio is given by:

$$G_t = \frac{C_{t+1}}{D_t}, \qquad D_t = 0.65\,C_t + 0.25\,C_{t-1} + 0.10\,C_{t-2}, \tag{11}$$

where C_t is the number of admissions in week t and D_t is the reporting scale, with $\sum_s {w_s}^2 = 0.495$. Only weeks with $D_t \geq 5$ were used for G. The renewal-type ratio converts single-week growth into a reproduction number through the generation interval (Wallinga and Lipsitch, 2007), and is given by:

$$R_{A,t} = (1 + r_t\theta)^{\alpha}, \qquad r_t = \frac{1}{7}\log\frac{C_{t+1} + 0.5}{C_t + 0.5}, \tag{12}$$

where r_t is the daily growth rate, and α and θ are the shape and scale of a gamma generation interval with mean 3.6 days and standard deviation 1.6 days, the household serial interval estimated by Cowling et al. (2009). Generation intervals with means of 2.2 and 4.3 days were used as sensitivity analyses. Because the scale of R_A is the single-week count, it was computed only for weeks with $C_t \geq 5$ as well as $D_t \geq 5$. R_A is a transformation of observed growth and not a posterior estimate of R_t (Cori et al., 2013); both statistics are therefore apparent quantities whose relationship to v_R was established by simulation (Section 6.8).

6.3. Expected loss and the balance scale

In routine surveillance the number of primary cases is unobserved, and secondary infections arise from the infection pressure $\Lambda_{t+1} = \sum_{s\geq 1} w_s I_{t+1-s}$ (Fraser, 2007; Cori et al., 2013). Applying the law of total variance to C_{t+1} given the history $\mathcal{F}_t$, with $m = \rho\mathbb{E}(\Lambda_{t+1} \mid \mathcal{F}_t)$, the relative prediction variance can be written as:

$$\frac{\mathrm{Var}(C_{t+1} \mid \mathcal{F}_t)}{m^2} = v_R + \frac{A}{m} + \mathrm{CV}^2(\Lambda_{t+1} \mid \mathcal{F}_t)\,(\bar{R}^2 + v_R), \tag{13}$$

where CV is the coefficient of variation. The last term is a further scale-invariant component that vanishes only when infection pressure is known. For a statistic X_t with mean $\bar{X}$, the relative loss $L_t = [(X_t - \bar{X})/\bar{X}]^2$ was modelled as:

$$\mathbb{E}(L_t \mid D_t) = a + \frac{b}{D_t}, \tag{14}$$

where, if a week were a single generation and D_t were free of noise, $a = v_R/\bar{R}^2$, $b = A/\bar{R}^2$ and $b/a = m_\times$. Binomial noise in the denominator, with relative variance $(1-\rho)\sum_s {w_s}^2/m$, increases the slope to approximately:

$$b \approx \frac{A + (1-\rho)(\bar{R}^2 + v_R)\sum_s {w_s}^2}{\bar{R}^2}. \tag{15}$$

Theoretical ranges of b/a for G used $\bar{R}$ equal to the mean of G in each specification, $v_R \in \{0.04, 0.08\}$, $k \in \{0.2, 0.5\}$ (Lloyd-Smith et al., 2005) and an admission probability per infection $\rho \in \{0.0056, 0.021\}$ (Tokars et al., 2018), with and without Eq. (15). The uncorrected balance scale over these scenarios was 21.3 to 49.2 cases.

6.4. Estimation and inference

Eq. (14) was fitted by feasible generalised least squares with weights $1/\hat{\mu}^2$, where $\hat{\mu}$ is the fitted mean from the previous iteration, iterated to convergence (Wooldridge, 2010). Observations from the same state are dependent over time, and observations from the same week share national influences. Uncertainty was therefore estimated by a two-way bootstrap that draws states and weeks independently with replacement and weights each observation by the product of the two draw counts (Cameron et al., 2008, 2011). Each specification used 2000 bootstrap replicates with its own seed (20261024 for R_A and 20261124 for G in the rule windows, 20261224 for G and 20261324 for R_A in the calendar windows), and uncertainty is reported as the point estimate followed by the 95% percentile interval, for example 0.0150 (95% CI: 0.0100, 0.0208).

The intercept was decomposed by refitting Eq. (14) twice. The first refit removed national-week effects by using deviations from the mean across states in each week, with a small-sample factor $n_w/(n_w - 1)$. The second removed additive week and state-season effects estimated by alternating projections. The national-week component is the difference between the unadjusted and week-adjusted intercepts, and the state-season component is the further change on adding state-season effects. The affine model was compared with a power law $\mathbb{E}(L_t \mid D_t) = c{D_t}^{-\gamma}$ by a weighted contrast of residual

sums of squares, calibrated by a residual bootstrap under the power law, and by leave-one-state-out and leave-one-season-out cross-validation. Power to detect a positive intercept was computed by simulating from the fitted affine model with intercepts up to the estimated value. Sensitivity analyses of b/a trimmed the most influential 1% of losses, omitted one season at a time, used Huber weights and removed national-week effects. Further analyses used the peak window, changed the generation interval and changed the denominator weights to a single week or three equal weeks.

6.5. Reporting completeness

Coverage-adjusted statistics were computed after dividing each count by the share of hospitals reporting in that state and week. The change in intercept, $\Delta a = a - a_{\text{adj}}$, was tested for equivalence to zero by two one-sided tests at the 5% level, using 90% intervals from a paired two-way bootstrap (Schuirmann, 1987). The margins were set at the intercept that a relative reporting volatility τ_0 would add under Proposition 2, and are given by:

$$\delta_G = {\tau_0}^2 \left(1 + \frac{v_R}{\bar{R}^2}\right), \qquad \delta_{R_A} = {\kappa_A}^2 {\tau_0}^2 \left(1 + \frac{v_R}{\bar{R}^2}\right), \tag{16}$$

where $\kappa_A = \mathrm{d}\log R_A / \mathrm{d}\log g$ propagates a perturbation of weekly growth g to R_A, $\bar{R}$ is the sample mean of each statistic, $v_R = 0.04$, and $\tau_0 = 0.10$ (primary) or 0.05. Because $\bar{R}$ is estimated, the margins expressed as shares of the intercept carry bootstrap intervals. Power was computed from a normal approximation. The interquartile range of $\log G$ and the standard deviation of G over May to October were compared between 2023, 2024 and 2025. As a second proxy, R_A was computed from weekly positive influenza tests in clinical laboratories for 44 states with available data, and the share of the hospital intercept removed by regressing hospital R_A on laboratory R_A was estimated with 500 two-way bootstrap replicates.

6.6. Seasons and pathogens

Season-specific intercepts were pooled with a DerSimonian–Laird random-effects model (DerSimonian and Laird, 1986), with a t interval on $S - 1$ degrees of freedom for the pooled intercept and a Q-profile interval for the between-season variance. Corollary 2 gives the lower bound on the precision of any between-season variance (Searle et al., 1992). Intercept ratios $q_1 = a_{(1,0,0)}/a_w$ and $q_2 = a_{(1/3,1/3,1/3)}/a_w$ compare a single-week and an equal three-week denominator with the weights of Eq. (11). They were computed for influenza in four windows and for COVID-19 and RSV admissions in two windows each. Null ranges came from simulated epidemic growth with reporting noise but no environmental variation, over three generation intervals, three admission delays and four noise levels. Each of the eight combinations was tested on five one-sided criteria ($q_1 < 1$, $q_2 > 1$, each ratio outside its null range and a positive excess over $v_R/\bar{R}^2$), and the 40 tests were adjusted by the Holm and Benjamini–Hochberg procedures (Holm, 1979; Benjamini and Hochberg, 1995).

6.7. Simulation of apparent statistics

The closed forms of Section 2 were checked by Monte Carlo with 20000 replications per point. In a first, generic design, apparent weekly statistics were simulated from a daily renewal model with $\bar{R} = 1.2$ and the generation interval above. Weekly R_t was drawn from a gamma distribution with variance v_R, independently across weeks, offspring were negative binomial and admissions were binomial with probability ρ_h after a gamma delay with mean 5.5 days and standard deviation 3.0 days, which was assumed. Each replicate contained 500 states with initial reporting scales log-uniform between 2 and 150, four warm-up weeks and six observation weeks. Seventeen scenarios crossed $v_R \in \{0.01, 0.02, 0.04, 0.08, 0.16\}$ with $k \in \{0.1, 0.35, 1\}$ at $\rho_h = 0.01$ and added $\rho_h \in \{0.005, 0.02\}$ at the baseline, each with 20 replicates. The noise-free limit of each intercept came from a mean-field run of 20000 states in which all count noise was replaced by expectations.

6.8. Two-step inversion

The empirical intercepts were mapped to v_R in two steps. In the first step, a data-matched version of the simulator gave the expected fitted intercept of each statistic as a function of v_R. Each simulated panel had the cross-section and length of the analysis panel, with 51 states and the four rule windows of 9, 12, 8 and 13 observation weeks. The season mean R_s was the renewal-type transform of the geometric-mean national weekly growth over each window, and initial admissions in each state equalled its observed admissions in the first window week. Weekly reproduction numbers were $R_{i,s,w} = R_s + \eta_{s,w} + \varepsilon_{i,s,w}$, where the national week shock η has variance $0.45\, v_R$ and is shared by all states, the state shock ε has variance $0.55\, v_R$, and values were truncated below at 0.05. The share of 0.45 was set between the national-week shares of 0.41 and 0.51 observed for the two statistics (Section 3.4); matching the simulated intraclass correlation and national-week share to the data is therefore a consistency check, and setting the share to zero or 0.65 changed the implied variance for R_A from 0.040 to 0.039 and 0.040 (Table S8). Transmission was a daily renewal process with negative binomial offspring ($k = 0.35$), admissions were binomial with probability $\rho_h = 0.01$ after the delay above, and both statistics were computed and filtered exactly as in the data, with the analyst's generation interval of mean 3.6 days. The design crossed $v_R \in \{0.02, 0.04, 0.06, 0.09\}$ with 200 panel replicates per value, and Eq. (14) was fitted to each replicate.

A calibration line $\mathbb{E}(\hat{a}) = \gamma_0 + \gamma_1 v_R$ was fitted by weighted least squares with weights equal to the inverse Monte Carlo variance of each mean, and the scale factor $\kappa = \mathbb{E}(\hat{a})/(v_R/\bar{R}^2)$ was computed with $\bar{R}$ the row-weighted true mean.

In the second step, the implied variance $v_R^{\text{impl}} = (a-\gamma_0)/\gamma_1$ was computed for each statistic. Its interval combined 2000 joint two-way bootstrap draws of the two empirical intercepts, which share state and week multiplicities, with independent draws of (γ_0, γ_1) from their Monte Carlo sampling distribution. Agreement between the two statistics was defined as non-rejection, that is, a 95% interval of their difference that contains zero. For the primary design, equivalence was also tested by two one-sided tests on the ratio of the implied variances with bounds of 0.80 and 1.25, using the 90% interval of the ratio draws. Sensitivity settings changed one element of the design at a time (Table S8). They removed the national shock or raised its share to 0.65, added first-order autocorrelation of 0.5 across weeks, set the true generation interval to a mean of 2.2 or 4.3 days while the analyst kept 3.6 days, and set the delay mean to 3.5 or 8 days. One further setting lowered k to 0.1, another replaced the transmission shock by a synchronous multiplicative shock to reported counts, and the generic design above served as a bridge. Four settings added at the final revision set the true generation interval to 2.9 days, the lower end of the interval of Cowling et al. (2009), set the first-order autocorrelation to 0.25, and drew weekly admissions from a gamma–Poisson mixture with a variance-to-mean ratio of 1.5 or 2.0 (Table S10). The independence of weekly shocks was checked with the pooled lag-1 autocorrelation of $x/\bar{x} - 1$ within state-season blocks of each statistic, with an interval from a state-cluster bootstrap of 1000 draws, and compared with its distribution over simulated panels at $v_R = 0.04$. The calibrated theoretical b/a used the same design over $v_R \in \{0.04, 0.08\}$, $k \in \{0.2, 0.5\}$ and $\rho_h \in \{0.0056, 0.021\}$, and the range of b/a across single replicates gave the distribution expected from one panel of this size. Simulation seeds were derived from 20261020 by a seed sequence indexed by setting, value of v_R and replicate block; the joint bootstrap and the calibration draws used 20261921 and 20261922.

Code availability

Code that reproduces every number, figure and table from the stored data, with fixed seeds and a single script that runs the full pipeline, is provided with this article in the `experiment/code` directory, together with a machine-readable record of each reported quantity. The analysis extends an earlier manuscript and package in the public repository https://github.com/wangchi2068/case-count-bottleneck. No archival DOI has been assigned to the code at the time of writing.

CRediT authorship contribution statement

Apevon Science: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

NHSN Weekly Hospital Respiratory Data are publicly available from https://data.cdc.gov/d/ua7e-t2fy; the version analysed was retrieved on 2026-10-20 and its SHA-256 checksum is recorded with the code. Clinical laboratory influenza data were obtained from the FluView clinical laboratory endpoint of the public Delphi Epidata interface (https://api.delphi.cmu.edu/epidata/). Derived panels used in the analyses are provided with the code.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools (Anthropic Claude) to restructure the analysis code, draft and revise the text, and draft the TikZ source of the conceptual schematic in Fig. 1A. All numerical results were produced by the deposited code and checked against its stored outputs. The authors reviewed and edited the content and take full responsibility for the content of the publication.

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This work re-analyses and extends an earlier manuscript on case volume and reproduction number error deposited in the repository cited under Code availability.

A. Proofs

A.1. Offspring and reporting model

The negative binomial distribution $\mathrm{NB}(r, p)$ with r successes and success probability p has mean $r(1-p)/p$ and variance $r(1-p)/p^2$. Setting $r = k$ and $p = k/(k + R_t)$ gives Eq. (1). A sum of n independent $\mathrm{NB}(k, p)$ variables is $\mathrm{NB}(nk, p)$, and binomial thinning of $\mathrm{NB}(r, k/(k + \mu))$ with retention probability ρ gives $\mathrm{NB}(r, k/(k + \rho\mu))$, which together give Eq. (2).

A.2. The period-specific target

Write c for an observed value of C_t. From Eq. (2), the log-likelihood of R_t given n is given by:

$$\ell(R_t) = \text{const} + nk \log \frac{k}{k + \rho R_t} + c \log \frac{\rho R_t}{k + \rho R_t}, \tag{A.1}$$

and the score can be written as:

$$\dot{\ell}(R_t) = \frac{c}{R_t} - \frac{(c + nk)\rho}{k + \rho R_t} = \mathcal{J}(R_t)\,(\hat{R}_t - R_t), \qquad \mathcal{J}(R_t) = \frac{n}{R_t/\rho + {R_t}^2/k}, \tag{A.2}$$

where $\mathcal{J}(R_t)$ is the Fisher information. The conditional mean of $\hat{R}_t = c/(\rho n)$ is R_t, and its conditional variance is $R_t/(\rho n) + {R_t}^2/(kn) = 1/\mathcal{J}(R_t)$. Because the score is proportional to $\hat{R}_t - R_t$, the Cramér–Rao bound is attained for every finite n. The distribution in Eq. (2) is a one-parameter exponential family in C_t with natural parameter $\log[\rho R_t/(k + \rho R_t)]$, which ranges over an open interval, so C_t is complete and sufficient and $\hat{R}_t$ is the uniformly minimum-variance unbiased estimator of R_t (Lehmann and Romano, 2005).

A.3. Proof of Proposition 1

Under H4, $\mathbb{E}(R_t \mid n) = \bar{R}$ and $\mathrm{Var}(R_t \mid n) = v_R$. By the law of total variance and Appendix A.2, the error variance is given by:

$$\mathrm{Var}(\hat{R}_t \mid n) = \mathbb{E}\left[\frac{R_t}{\rho n} + \frac{{R_t}^2}{kn}\right] + \mathrm{Var}(R_t) = v_R + \frac{\bar{R}}{\rho n} + \frac{\bar{R}^2 + v_R}{kn}, \tag{A.3}$$

and substituting $m = \rho n$ gives Eq. (4). Parts (i) and (ii) follow by taking limits. For part (iii), $[V(m) - V(2m)]/V(m) = [A/(2m)]/(v_R + A/m) = 1/[2(1 + m/m_\times)]$, which equals one quarter at $m = m_\times$. For part (iv), let $\tilde{\theta}$ be any estimator from data on period t with $\mathbb{E}(\tilde{\theta} \mid R_t) = R_t$. Then $\mathbb{E}(\tilde{\theta} - \bar{R})^2 = \mathbb{E}[\mathrm{Var}(\tilde{\theta} \mid R_t)] + v_R \geq \mathbb{E}[1/\mathcal{J}(R_t)] + v_R = V$, with equality for $\hat{R}_t$. For one-step prediction of $y = C_{t+1}/m$ when R_{t+1} is independent of the history, $\mathbb{E}(\hat{y} - y)^2 = (\hat{y} - \mathbb{E}y)^2 + \mathrm{Var}(y) \geq V$, with equality at $\hat{y} = \bar{R}$.

The marginal effects of coverage and completeness are $-\partial V/\partial \log n = A/m$ and $-\partial V/\partial \log \rho = \bar{R}/m$, so their ratio is $A/\bar{R}$. If R_t follows a stationary first-order autoregressive process with variance v_R and autocorrelation ϕ, the best linear predictor of R_{t+1} given R_t has error variance $(1 - \phi^2)v_R$ (Hamilton, 1994). For J jurisdictions with common variance v_R and mean pairwise correlation $\bar{c}$, the variance of their average is $J^{-2}[Jv_R + J(J-1)\bar{c}v_R]$, which gives Eq. (5).

A.4. Proof of Proposition 2

Write $\sigma^2 = \mathrm{Var}(p_t) = \tau^2\rho^2$. Given (n, R_t, p_t), C_t has mean $p_t n R_t$ and variance $p_t n R_t + {p_t}^2 n {R_t}^2/k$. Averaging over p_t, the conditional variance given (n, R_t) is given by:

$$\mathrm{Var}(C_t \mid n, R_t) = \rho n R_t + \frac{(\rho^2 + \sigma^2) n {R_t}^2}{k} + \sigma^2 n^2 {R_t}^2. \tag{A.4}$$

The conditional mean is $\rho n R_t$, so $\hat{R}_t$ remains unbiased for $\bar{R}$. Averaging over R_t and dividing by $(\rho n)^2$ gives:

$$V_H(m) = v_R + \tau^2(\bar{R}^2 + v_R) + \frac{\bar{R}}{m} + \frac{\rho(1 + \tau^2)(\bar{R}^2 + v_R)}{km}, \tag{A.5}$$

which is Eq. (6). Writing $U = (1 + \tau^2)(\bar{R}^2 + v_R)$, the limiting error variance is $v_R + v_H = U - \bar{R}^2$ and $A_H = \bar{R} + \rho U/k$, so the mean and variance of $\hat{R}_t$ depend on v_R and τ only through U. The inequality $m_{\times,H} < m_\times$ is equivalent to $\rho v_R/k < A$, which holds because $\bar{R}^2 + v_R > v_R$.

A.5. *Proof of Proposition 3*

Conditional on $(R_1, \dots, R_T)$, counts in different periods are independent. With $\bar{R}_T = T^{-1}\sum_t R_t$, the variance of the mean estimate is given by:

$$\mathrm{Var}\Big(\frac{1}{T}\sum_{t=1}^{T}\hat{R}_t\Big) = \mathrm{Var}(\bar{R}_T) + \frac{A}{Tm}, \tag{A.6}$$

where $\mathrm{Var}(\bar{R}_T) = v_R/T$ for independent periods, giving Eq. (7). For a stationary first-order autoregressive process with $\mathrm{Cov}(R_s, R_t) = v_R\phi^{|s-t|}$, $T^2\mathrm{Var}(\bar{R}_T) = v_R[T + 2\sum_{h=1}^{T-1}(T-h)\phi^h]$, and summing the geometric series gives Eq. (9) (Hamilton, 1994). Substituting $T = \mathcal{B}/(c_0 + c_1 m)$ into Eq. (7) gives the objective:

$$f(m) = \frac{(c_0 + c_1 m)(v_R + A/m)}{\mathcal{B}} = \frac{c_0 v_R + c_1 A + c_1 v_R m + c_0 A/m}{\mathcal{B}}, \tag{A.7}$$

which is strictly convex for $c_0 > 0$, with $f'(m) = (c_1 v_R - c_0 A/m^2)/\mathcal{B}$. The unique stationary point is $m^{*2} = c_0 A/(c_1 v_R) = (c_0/c_1)m_\times$, which is Eq. (8).

A.6. *Proof of Corollary 1*

Write $U_0 = \bar{R}^2 + v_R$. From Eq. (6), the gains from proportional increases in coverage and completeness are given by:

$$-\frac{\partial V_H}{\partial \log n} = \frac{\bar{R}}{\rho n} + \frac{(1+\tau^2)U_0}{kn}, \qquad -\frac{\partial V_H}{\partial \log \rho} = \frac{\bar{R}}{\rho n} - \rho\frac{\mathrm{d}\tau^2}{\mathrm{d}\rho}\,U_0\left(1 + \frac{1}{kn}\right). \tag{A.8}$$

If $\mathrm{d}\tau^2/\mathrm{d}\rho = 0$, the first gain exceeds the second by $(1+\tau^2)U_0/(kn) > 0$. Under binomial sampling of ν units, $\tau^2 = (1-\rho)/[\rho(\nu+1)]$ and $\rho\,\mathrm{d}\tau^2/\mathrm{d}\rho = -1/[\rho(\nu+1)]$, and the second gain minus the first is

$$\frac{U_0}{kn}\left[\frac{kn+\rho}{\rho(\nu+1)} - 1\right], \tag{A.9}$$

which is positive if and only if $kn > \rho\nu$, that is $m > \rho^2\nu/k$.

A.7. *Renewal formulation and denominator noise*

Let $\bar{\Lambda} = \mathbb{E}(\Lambda_{t+1} \mid \mathcal{F}_t)$ and $m = \rho\bar{\Lambda}$. Conditioning on (Λ_{t+1}, R_{t+1}) and applying the law of total variance, the prediction variance is given by:

$$\mathrm{Var}(C_{t+1} \mid \mathcal{F}_t) = mA + \rho^2\bar{\Lambda}^2\left[v_R + \mathrm{CV}^2(\Lambda_{t+1} \mid \mathcal{F}_t)\,(\bar{R}^2 + v_R)\right], \tag{A.10}$$

and dividing by m^2 gives Eq. (13). For Eq. (15), write $D_t = m(1+\delta)$ with $\mathbb{E}\delta = 0$ and $\mathbb{E}\delta^2 = c_w/m$, where $c_w = (1-\rho)\sum_s {w_s}^2$, and let $Y = C_{t+1}/m$ be independent of δ with mean $\bar{R}$ and variance $v_R + A/m$. Expanding $G_t = Y(1+\delta)^{-1}$ to second order gives:

$$\mathbb{E}(G_t) = \bar{R}\left(1 + \frac{c_w}{m}\right) + O(m^{-2}), \qquad \mathrm{Var}(G_t) = v_R + \frac{A + (\bar{R}^2 + 3v_R)c_w}{m} + O(m^{-2}), \tag{A.11}$$

and dividing by $[\mathbb{E}(G_t)]^2 = \bar{R}^2(1 + 2c_w/m) + O(m^{-2})$ gives a relative variance of $v_R/\bar{R}^2 + [A + (\bar{R}^2 + v_R)c_w]/(\bar{R}^2 m)$ to first order in $1/m$, which is Eq. (15).

B. Extensions

B.1. *Precision of the environmental variance*

Proposition 4 (Precision of environmental variance)**.** *Suppose that $R_1, \dots, R_T$ are independent gamma variables with known mean $\bar{R}$ and variance v_R, and let $\xi = \bar{R}^2/v_R$. Any unbiased estimator of v_R based on the counts from the T periods has relative standard error at least*

$$\left[T\xi\,(\xi\psi'(\xi) - 1)\right]^{-1/2}, \tag{B.1}$$

where ψ' is the trigamma function. As $\xi \to \infty$, the bound tends to the Gaussian value $\sqrt{2/T}$.

Proof. The counts depend on v_R only through $R_{1:T}$, so by the data processing inequality the Fisher information for v_R in the counts cannot exceed that in $R_{1:T}$. With the mean fixed, the gamma shape is ξ and the scale is $v_R/\bar{R}$. The information for ξ in one observation is $\psi'(\xi) - 1/\xi$, and $\mathrm{d}\xi/\mathrm{d}v_R = -\xi/v_R$, so the information for v_R in T periods is $T\xi(\xi\psi'(\xi) - 1)/{v_R}^2$. The Cramér–Rao inequality gives Eq. (B.1). The limit follows from $\psi'(\xi) = 1/\xi + 1/(2\xi^2) + O(\xi^{-3})$. □

At the baseline of Table S7 ($\bar{R} = 1.15$, $v_R = 0.04$), $\xi = 33.06$ and $\xi(\xi\psi'(\xi) - 1) = 0.505$, so the relative standard error is at least 0.812 with 3 periods, 0.445 with 10 periods and 0.199 with 50 periods. The constant 0.505 is specific to this baseline, and Eq. (B.1) is the general form.

B.2. Budget threshold with autocorrelated environments

With autocorrelation ϕ, the environmental term of Proposition 3 is $v_R e(T)$ with $e(T) = S(\phi, T)/T^2$. Define $g_T = -T^2 e'(T)$, treating T as continuous, so that $g_T = 1$ for independent periods, and let ${m_0}^* = \sqrt{(c_0/c_1)m_\times}$ be the optimum of Eq. (8).

Proposition 5 (Budget threshold)**.** *Under the budget constraint $\mathcal{B} = T(c_0 + c_1 m)$, the optimal case volume per period satisfies $m^* = {m_0}^*(g_{T^*})^{-1/2}$, where T^* is the optimal number of periods. If g_T increases in T from $g_1 < 1$ towards $(1+\phi)/(1-\phi) > 1$, there is a unique $T^\dagger$ with $g_{T^\dagger} = 1$, and m^* exceeds ${m_0}^*$ when $T^* < T^\dagger$ and falls below it when $T^* > T^\dagger$. The two regimes meet at the budget*

$$\mathcal{B}^\dagger = T^\dagger \left(c_0 + c_1 {m_0}^*\right), \tag{B.2}$$

which is expressed in units of the cost per reported case when $c_1 = 1$.

Proof. Writing the sampling term as $Ac_1/(\mathcal{B} - c_0 T)$ and differentiating $v_R e(T) + Ac_1/(\mathcal{B} - c_0 T)$ with respect to T gives the first-order condition $v_R g_T = c_0 A/(c_1 m^2)$, so $m^{*2} = {m_0}^{*2}/g_{T^*}$. The ordering of m^* and ${m_0}^*$ follows from the monotonicity of g_T, and at $T^* = T^\dagger$ the budget constraint gives Eq. (B.2). □

For $\phi = 0.8$, g_T increases in T and $T^\dagger = 4.484$. With $c_0/c_1 = 100$ and the baseline balance scale, ${m_0}^* = 72.86$ and $\mathcal{B}^\dagger = 775.0$; over the latent balance scales of 21.3 to 49.2 cases used in Section 6.3, $\mathcal{B}^\dagger$ ranges from 655.3 to 762.8 (Table S7). The calculation treats T as continuous and the environmental process as stationary, so it applies within a season and not across seasons with different mean transmission.

B.3. Between-season variance

Corollary 2 (Season-level bound)**.** *Suppose that season effects $\eta_1, \ldots, \eta_S$ are independent Gaussian variables with unknown mean and variance v_S, and that the counts depend on v_S only through the season effects. Any unbiased estimator of v_S based on the counts has relative standard error at least $\sqrt{2/(S-1)}$, which is 1.00 for $S = 3$ and 0.816 for $S = 4$.*

Proof. Let δ be an unbiased estimator based on the counts. Because the conditional distribution of the counts given $\eta_{1:S}$ does not depend on v_S, $\mathbb{E}(\delta \mid \eta_{1:S})$ is a function of the season effects that is unbiased for v_S, and its variance does not exceed that of δ. The sample mean and sample variance of $\eta_{1:S}$ are complete and sufficient, so the sample variance is the minimum-variance unbiased estimator of v_S, with variance $2{v_S}^2/(S-1)$ (Lehmann and Romano, 2005; Searle et al., 1992). □

The bound uses $S-1$ degrees of freedom because the mean across seasons is estimated; the corresponding bound with a known mean, $\sqrt{2/S}$, is smaller (Table S7). Season-specific intercepts for three and four seasons are reported separately in Table S5.

B.4. Testing whether transmission exceeds one

For $R > R'$, the likelihood ratio of Eq. (2) increases in c because $R/(k+\rho R)$ increases in R, so the family has a monotone likelihood ratio in C_t. The test that rejects $R_t \le 1$ when $C_t \ge c_\alpha$ therefore has a power function that does not decrease in R_t, and by the Karlin–Rubin theorem it is uniformly most powerful at its attained level $\alpha^* = P_{R_t=1}(C_t \ge c_\alpha) \le \alpha$ (Karlin and Rubin, 1956; Lehmann and Romano, 2005). Because C_t is discrete, α^* is generally below the nominal level; optimality holds at α^*, not at α, unless the test is randomised or a mid-p value is used.

B.5. Shrinkage as an ideal bound

Part (iv) of Proposition 1 excludes estimators that are biased given R_t. Consider $\hat{\theta}_\lambda = (1-\lambda)\hat{R}_t + \lambda\mu_0$ for a fixed benchmark μ_0 with $\Delta = \mu_0 - \bar{R}$. Its mean squared error for $\bar{R}$ is given by:

$$\mathrm{MSE}(\lambda) = (1-\lambda)^2 V + \lambda^2 \Delta^2, \tag{B.3}$$

where V is given by Eq. (4). The minimum is at $\lambda^* = V/(V+\Delta^2)$, where $\mathrm{MSE}(\lambda^*) = V\Delta^2/(V+\Delta^2)$, which is below v_R if and only if $\Delta^2 < v_R(1 + m/m_\times)$. The weight λ^* depends on the unknown $\bar{R}$ and V, so this is an ideal bound and not a feasible estimator; an empirical Bayes or hierarchical estimator estimates the weight from the data and does not attain it (Stoner et al., 2019).

C. Supplementary figure

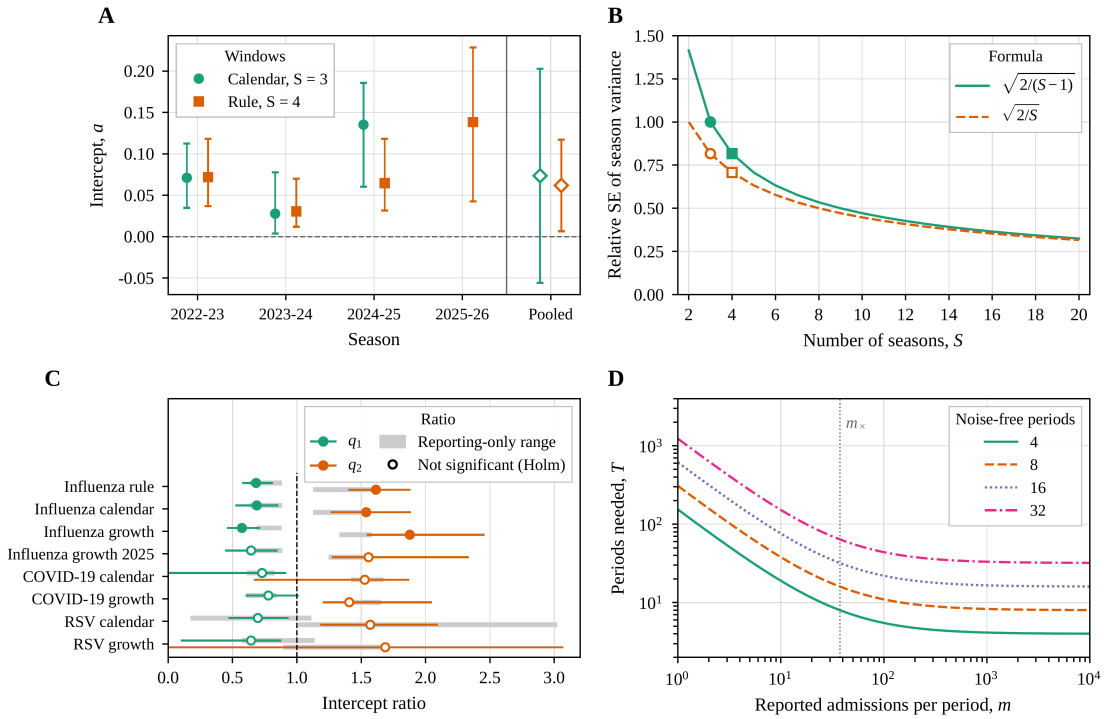


Fig. 7. Between-season uncertainty, intercept ratios across pathogens and number of periods. (A) Season-specific intercepts of the weighted growth ratio in calendar windows and in rule-defined early-growth windows, with two-way bootstrap 95% intervals; open diamonds show random-effects pooled estimates. (B) Lower bound on the relative standard error of an estimated between-season variance against the number of seasons, compared with the values reported in the earlier analysis. (C) Intercept ratios for a single-week denominator and an equal three-week denominator relative to the original weights; grey bars give the ranges expected from reporting noise and epidemic growth without environmental variation, and filled markers denote one-sided tests that remain significant after Holm correction. (D) Number of periods needed to match the precision of a given number of noise-free periods, by reported admissions per period; the panel is illustrative and is superseded by the empirical nomogram in Fig. 6A.

D. Supplementary tables

Tables S1–S7 report the estimates that underlie Figs. 4–7. Table S8 gives the implied environmental variance under each setting of the inversion (Fig. 3), and Table S9 lists the specifications used across the rounds of this analysis. Table S4 compares the voluntary reporting period with the same calendar months of the mandatory years. All values were generated from the stored outputs of the analysis code.

Table S1. Intercept a, slope b and balance ratio b/a of the relative loss by specification, with the two-way bootstrap 95% confidence interval of each estimate.

Specification	N	Weeks	$\bar{x}$	a [95% CI]	b [95% CI]	b/a [95% CI]	SD ratio
R_A, rule windows ($S = 4$)	1748	42	1.206	0.0150 [0.0100, 0.0208]	0.560 [0.367, 0.769]	37.4 [19.1, 71.0]	1.95
G, rule windows ($S = 4$)	1756	42	1.724	0.0840 [0.0498, 0.1248]	1.971 [0.959, 3.098]	23.5 [8.4, 59.7]	2.29
G, calendar windows ($S = 3$)	1267	29	1.671	0.0991 [0.0569, 0.1382]	2.243 [1.060, 3.559]	22.6 [9.0, 52.2]	4.68
R_A, calendar windows ($S = 3$)	1263	29	1.190	0.0202 [0.0097, 0.0306]	0.516 [0.279, 0.839]	25.6 [10.2, 79.6]	3.46

Iterated FGLS fit of $L = ((x - \bar{x})/\bar{x})^2$ on $a + b/D$; 95% percentile intervals from the two-way (state × week) pigeonhole bootstrap, $B = 2000$, $\bar{x}$ re-estimated in every replicate. SD ratio: bootstrap SD of a under two-way versus state-only resampling. 51 jurisdictions throughout.

Table S2. Decomposition of the intercept and comparison of the affine model with a power law, with the 95% confidence interval of each intercept.

Specification	a total	a after week effects	a after week and state-season effects	Week share	ICC	Power exponent	p	CV folds (state; season)
R_A, rule windows ($S = 4$)	0.0150 [0.0100, 0.0208]	0.0088 [0.0064, 0.0124]	0.0083 [0.0064, 0.0141]	0.41	0.25	0.38	0.11	26/51; 2/4
G, rule windows ($S = 4$)	0.0840 [0.0508, 0.1226]	0.0415 [0.0306, 0.0586]	0.0394 [0.0293, 0.0614]	0.51	0.31	0.35	0.44	19/51; 0/4
G, calendar windows ($S = 3$)	0.0991 [0.0583, 0.1399]	0.0473 [0.0337, 0.0676]	0.0450 [0.0335, 0.0932]	0.52	0.29	0.27	0.42	28/51; 3/3

Intervals are from a two-way bootstrap with $B = 1000$; the draws are separate from those of Table S1, which explains small differences. The third column removes national week effects from the statistic before the fit, and the fourth also removes state-season effects. The week share equals one minus the ratio of the third to the second column. The ICC is the share of the variance of the statistic that lies between weeks. The power exponent is c in $L = \alpha D^{-c}$, and p is from a parametric-bootstrap test of the affine model against this power law with 200 null draws. CV folds counts the leave-one-state-out and leave-one-season-out folds in which the affine model predicts better. A power analysis found no fraction of the estimated a (≤ 1) detectable with 80% power.

Table S3. Equivalence tests for the contribution of facility-reporting completeness to the intercept, with the paired bootstrap 90% confidence interval of each change.

Specification	$10^3 \Delta a$ [90% CI]	$10^3 \delta_{0.10}$	Equiv.	$10^3 \delta_{0.05}$	Equiv.	Seasons equiv.
R_A, rule windows ($S = 4$)	0.29 [−0.22, 0.85]	2.55	yes	0.64	no	4/4
G, rule windows ($S = 4$)	0.37 [−2.83, 2.81]	10.13	yes	2.53	no	3/4
G, calendar windows ($S = 3$)	−0.38 [−3.55, 2.24]	10.14	yes	2.54	no	2/3

Δa is the intercept of the statistic minus that of the statistic recomputed from counts divided by the share of hospitals reporting, with a paired two-way bootstrap ($B = 2000$). Margins $\delta_{\tau_0} = \kappa^2 \tau_0^2 (1 + v_R/\bar{R}^2)$ were set from a reporting-volatility threshold and the sample mean $\bar{R}$ of each statistic ($\kappa = 1$ for G, $v_R = 0.04$). Equivalence is declared when the 90% interval lies inside $(-\delta, \delta)$ (TOST, $\alpha = 0.05$). The column of equivalent seasons counts seasons equivalent at $\tau_0 = 0.10$.

Table S4. Calendar-matched comparison of the voluntary-reporting period (May–October).

Year	Reporting	N	SD of G	IQR of $\log G$
2023	mandatory	773	0.62 [0.40, 1.00]	0.48 [0.40, 0.61]
2024	voluntary	420	1.82 [0.42, 4.10]	0.76 [0.48, 1.04]
2025	mandatory	841	0.39 [0.33, 0.44]	0.48 [0.37, 0.58]

State-weeks with $D_t \geq 5$ whose week ends between 1 May and 31 October; 95% two-way bootstrap intervals, $B = 1000$. The FGLS intercept is not stable in the voluntary period (negative point estimate), so robust spread measures are reported. Laboratory proxy (Delphi FluView clinical laboratory data, 1224 state-weeks): share of the R_A intercept shared with laboratory growth 0.23 [0.043, 0.42]; correlation 0.31 [0.19, 0.41].

Table S5. Season-specific intercepts, each with its bootstrap 95% confidence interval, and between-season heterogeneity, reported separately for $S = 4$ and $S = 3$.

Season	R_A, rule windows ($S = 4$)	G, rule windows ($S = 4$)	G, calendar windows ($S = 3$)
2022-23	0.0137 [0.0066, 0.0213]	0.0718 [0.0368, 0.1181]	0.0711 [0.0347, 0.1124]
2023-24	0.0062 [0.0025, 0.0144]	0.0305 [0.0119, 0.0700]	0.0277 [0.0037, 0.0776]
2024-25	0.0127 [0.0057, 0.0241]	0.0647 [0.0315, 0.1181]	0.1352 [0.0602, 0.1857]
2025-26	0.0225 [0.0087, 0.0352]	0.1383 [0.0426, 0.2285]	–
a_{RE} [95% CI]	0.0125 [0.0022, 0.0227]	0.0619 [0.0066, 0.1171]	0.0735 [−0.0558, 0.2028]
Q (p)	5.71 (0.126)	6.58 (0.087)	8.92 (0.012)
$10^3\tau^2$ [Q-profile]	0.019 [0.000, 0.599]	0.620 [0.000, 26.756]	2.077 [0.127, 114.859]
I^2	0.47	0.54	0.78

Season intercepts use the season's own mean; 95% two-way bootstrap intervals, $B = 1000$. DerSimonian–Laird random effects with a t_{S-1} interval; Q-profile 95% interval for τ^2. The relative standard error of a between-season variance estimated from S seasons is at least $\sqrt{2/(S-1)}$: 1.00 for $S = 3$ and 0.82 for $S = 4$.

Table S6. Cross-pathogen intercept ratios against the pure-reporting growth null, with Holm-adjusted p-values.

Pathogen, window	N	q_1 [95% CI]	q_2 [95% CI]	Null q_1	Null q_2	$q_1 < 1$	$q_2 > 1$	q_1 out	q_2 out	Excess > 0
Influenza, rule	1756	0.68 [0.58, 0.81]	1.61 [1.41, 1.87]	0.63–0.89	1.13–1.58	0.020	0.034	1.000	1.000	0.034
Influenza, calendar	1267	0.69 [0.53, 0.85]	1.54 [1.27, 1.88]	0.63–0.89	1.13–1.58	0.020	0.020	1.000	1.000	0.020
Influenza, growth	2707	0.57 [0.47, 0.71]	1.88 [1.55, 2.45]	0.68–0.88	1.33–1.58	0.034	0.020	1.000	1.000	0.020
Influenza, growth, 2025-26	663	0.64 [0.45, 0.84]	1.56 [1.28, 2.33]	0.66–0.89	1.25–1.57	0.116	0.075	1.000	1.000	0.046
COVID-19, calendar	1479	0.73 [n.d., 0.91]	1.53 [0.67, 1.87]	0.61–0.83	1.42–1.67	0.287	0.616	1.000	1.000	1.000
COVID-19, growth	2862	0.78 [0.61, 1.01]	1.41 [1.21, 2.04]	0.60–0.84	1.40–1.66	0.616	0.154	1.000	1.000	1.000
RSV, calendar	494	0.70 [0.48, 0.93]	1.57 [1.19, 2.09]	0.17–1.11	1.01–3.02	0.154	0.228	1.000	1.000	0.200
RSV, growth	1073	0.64 [0.11, 0.87]	1.69 [n.d., 3.06]	0.57–1.14	0.89–1.69	0.182	0.660	1.000	1.000	1.000

Here $q_1 = a_{\text{1 week}}/a_{\text{original}}$ and $q_2 = a_{\text{equal 3 weeks}}/a_{\text{original}}$, with intervals from a shared two-way bootstrap ($B = 2000$). Null ranges come from a growth-null simulation without transmission heterogeneity over three generation intervals, three admission delays and four reporting-noise levels; the rule-window row uses the influenza calendar range. The last five columns give one-sided bootstrap p-values, Holm-adjusted over all 40 tests, and the columns marked out test whether the ratio lies outside the null range. Excess denotes $a - v_R/G^2$ with $v_R = 0.04$. A lower bound marked n.d. is not defined because some bootstrap replicates gave a negative intercept in the ratio, so the ratio has no positive lower limit; the upper limit is shown as estimated.

Table S7. Theory quantities recomputed from the closed forms and compared with an earlier version of this analysis.

Quantity	Earlier	Recomputed	Status
$m_\times$	53.1	53.08	reproduced
v_H ($\tau = 0.15$)	0.0307	0.0307	reproduced
$m_{\times,H}$ ($\tau = 0.15$)	30.4	30.36	reproduced
m_0^* ($c_0/c_1 = 100$)	72.9	72.86	reproduced
$T^\dagger$ ($\phi = 0.8$)	4.48	4.484	reproduced
$B^\dagger$	775	775.0	reproduced
$B^\dagger$ ($m_\times = 21.3$)	655	655.3	reproduced
$B^\dagger$ ($m_\times = 49.2$)	763	762.8	reproduced
Reporting lever term ($\times 10^{-3}$)	7.85	7.88	corrected
Lever total ($\times 10^{-3}$)	22.5	22.57	corrected
Fisher constant	0.505	0.505	reproduced
Relative SE, $S = 3$	0.82	1.000	corrected
Relative SE, $S = 4$	0.71	0.816	corrected

Baseline $\bar{R} = 1.15$, $v_R = 0.04$, $k = 0.35$, $\rho = 0.25$. The earlier relative standard errors for $S = 3, 4$ equal $\sqrt{2/S}$; the stated bound $\sqrt{2/(S-1)}$ gives the recomputed values. The lever terms differ by rounding.

Table S8. Implied environmental variance v_R^{impl} from the two-step inversion under each simulation setting, with the 95% confidence interval of each estimate.

Setting	v_R^{impl} from R_A	v_R^{impl} from G	κ_{R_A}	κ_G	Ratio R_A/G	Agree
Data-matched design (M0)	0.040 [0.026, 0.056]	0.043 [0.025, 0.064]	0.516	2.729	0.93 [0.78, 1.18]	yes
No national shock	0.039 [0.026, 0.055]	0.042 [0.024, 0.062]	0.523	2.775	0.94 [0.79, 1.18]	yes
National share 0.65	0.040 [0.026, 0.056]	0.043 [0.025, 0.065]	0.510	2.670	0.92 [0.77, 1.16]	yes
AR(1), $\phi = 0.5$	0.027 [0.017, 0.040]	0.025 [0.013, 0.039]	0.780	4.699	1.08 [0.88, 1.43]	yes
True GI mean 2.2 d	0.013 [0.008, 0.019]	0.014 [0.008, 0.021]	1.287	6.991	0.94 [0.77, 1.20]	yes
True GI mean 4.3 d	0.058 [0.038, 0.081]	0.063 [0.037, 0.095]	0.380	1.977	0.92 [0.77, 1.16]	yes
Delay mean 3.5 d	0.038 [0.025, 0.053]	0.041 [0.024, 0.061]	0.535	2.790	0.93 [0.77, 1.16]	yes
Delay mean 8 d	0.046 [0.030, 0.064]	0.047 [0.027, 0.069]	0.454	2.483	0.98 [0.82, 1.22]	yes
$k = 0.1$	0.040 [0.026, 0.055]	0.042 [0.025, 0.062]	0.517	2.740	0.94 [0.79, 1.18]	yes
Synchronous count shock	0.019 [0.006, 0.035]	0.029 [0.012, 0.048]	1.041	4.553	0.67 [0.39, 0.90]	no
Generic design (bridge)	0.043 [0.029, 0.059]	0.044 [0.027, 0.063]	0.507	2.734	0.97 [0.83, 1.18]	–

Each setting changes one element of the data-matched design (Section 6.8); the analyst's generation interval stays at a mean of 3.6 days throughout. Intervals combine 2000 joint two-way bootstrap draws of the two empirical intercepts with draws of the calibration line from its Monte Carlo sampling distribution. $\kappa = \mathbb{E}(\hat{a})/(v_R/\bar{R}^2)$ at the grid value of v_R nearest to the implied variance. Agree marks settings in which the 95% interval of the difference between the two implied variances contains zero; the generic design of the first simulation is shown as a bridge and is not counted among the ten settings.

Table S9. Specifications used across the rounds of this analysis, with the 95% confidence interval of each estimate on the data version of 2026-10-20.

Round and specification	Role	a [95% CI]	b/a [95% CI]
Round 1: G, three calendar windows ($S = 3$)	Primary in round 1; secondary from round 2	0.0991 [0.0569, 0.1382]	22.6 [9.0, 52.2]
Round 2, rule version 1: windows ending at the season maximum	Abandoned (four weeks in 2024–25); no estimates retained	–	–
Round 2, rule version 2 (2026-10-20): G, rule windows ($S = 4$)	Secondary	0.0840 [0.0498, 0.1248]	23.5 [8.4, 59.7]
Round 2, rule version 2: R_A, rule windows ($S = 4$)	Primary from round 2	0.0150 [0.0100, 0.0208]	37.4 [19.1, 71.0]
Round 2: R_A, calendar windows ($S = 3$)	Sensitivity	0.0202 [0.0097, 0.0306]	25.6 [10.2, 79.6]
Round 3: two-step inversion	Calibration of the round-2 fits; no refit	–	–

All estimates were refitted on the same data version with two-way bootstrap intervals ($B = 2000$). The renewal-type ratio became the primary statistic in round 2 because its relationship to R_t is explicit through the generation interval; the weighted growth ratio is retained for continuity. The point estimate of each specification lies inside the interval of the other specification of the same statistic.

Table S10. Lag-1 autocorrelation, and implied environmental variance and b/a in the added inversion settings.

A. Lag-1 autocorrelation

Setting	r_1 of R_A	Data pct.	r_1 of G	Data pct.
Data (rule windows)	-0.095 [-0.168, -0.019]	–	0.123 [0.039, 0.201]	–
Data-matched design (M0)	-0.074 [-0.151, 0.014]	31.0	0.114 [0.025, 0.220]	62.0
AR(1), $\phi = 0.25$	0.007 [-0.098, 0.118]	3.0	0.193 [0.097, 0.304]	10.0
AR(1), $\phi = 0.5$	0.072 [-0.039, 0.229]	0.0	0.258 [0.146, 0.390]	0.0
True GI mean 2.2 d	0.099 [-0.002, 0.223]	0.0	0.251 [0.156, 0.363]	0.0
True GI mean 2.9 d	0.006 [-0.083, 0.120]	2.5	0.183 [0.091, 0.296]	9.0
True GI mean 4.3 d	-0.134 [-0.206, -0.030]	80.0	0.051 [-0.031, 0.161]	92.0
Admissions VMR 1.5	-0.170 [-0.233, -0.096]	97.5	-0.014 [-0.092, 0.076]	100.0
Admissions VMR 2.0	-0.183 [-0.247, -0.104]	99.0	-0.041 [-0.104, 0.035]	100.0
Synchronous count shock	-0.146 [-0.225, -0.076]	86.0	0.041 [-0.033, 0.116]	99.0

B. Added settings (b/a pooled at $v_R = 0.04$)

Setting	v_R^{impl} from R_A	v_R^{impl} from G	b/a of R_A	b/a of G
Data-matched design (M0)	0.040 [0.026, 0.056]	0.043 [0.025, 0.064]	28.0	16.2
True GI mean 2.9 d	0.026 [0.017, 0.036]	0.027 [0.016, 0.040]	18.2	10.4
Admissions VMR 1.5	0.039 [0.026, 0.055]	0.044 [0.026, 0.066]	68.0	40.2
Admissions VMR 2.0	0.039 [0.026, 0.055]	0.045 [0.027, 0.066]	81.0	48.4
AR(1), $\phi = 0.25$	0.033 [0.021, 0.047]	0.033 [0.019, 0.049]	23.4	12.6

r_1 is the pooled lag-1 autocorrelation of the relative deviation $x/\bar{x}_{\text{block}} - 1$ within state-season blocks. The interval for the data comes from a state-cluster bootstrap with 1000 draws; simulated rows give the mean and the 2.5% to 97.5% range of r_1 over panels at $v_R = 0.04$ (200 panels per value of v_R for the added settings, 100 for the others), and Data pct. is the percentile of the empirical value in that distribution. The analyst's generation interval stays at a mean of 3.6 days; 2.9 and 4.3 days are the ends of the 95% interval for the serial interval of influenza in Cowling et al. (2009). Overdispersed admissions use a gamma–Poisson mixture on weekly counts.

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